

Analysis of Diffusion Properties in Generalized Feistel Ciphers Under Multidimensional Linear Cryptanalysis

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Abstract. This paper presents a unified framework for generic attacks on Generalized Feistel Ciphers, with a primary focus on Type 1, Type 2, and unbalanced contracting (U-Type 1) Feistel constructions with non-invertible round functions. In [OBR23], authors reveal a class of vulnerabilities exploitable via key independent multidimensional linear trails for Feistel Ciphers, yielding efficient generic distinguishing and key-recovery attacks. We extend the work of [OBR23] by formalizing the application of generic multidimensional linear cryptanalysis to Generalized Feistel Ciphers. In this way, we improve upon existing results by extending the maximum number of rounds for the generic distinguishing attack to $t^2 + 2t - 1$ for Type 1 and U-Type 1, and to $2t + 3$ for Type 2. Moreover, we have the maximum number of rounds for generic key recovery attacks on (U)-Type 1 as $t^2 + 3t - 2$ and Type 2 as $4t$. To the best of our knowledge, these findings yield the best results for the maximum number of rounds in key recovery attacks on the corresponding GFC. We further demonstrate the branch-permutation-independence of these attacks, showing that changing internal permutations does not affect the applicability, complexity, or maximum number of rounds of generic attacks. The effectiveness of our attacks is validated through experiments on the first-round AES candidate CAST-256 and the MPC-friendly block cipher GMiMC. Both theoretical and experimental results confirm that our proposed branch-permutation-independent generic attacks enhance the maximum number of rounds for generic attacks for GFC and reduce complexity across various interesting cases.

Keywords: Generic Attack · Multidimensional Linear Cryptanalysis · Generalized Feistel Ciphers · Diffusion Property · CAST-256 · GMiMC.

1 Introduction

Generic attacks on symmetric primitives are a type of cryptanalytic approach that does not exploit specific structural weaknesses in a given cipher but instead applies general techniques that can be used across a wide range of cryptographic algorithms. This technique ensures that the security evaluation reflects the inherent resilience of the primitive rather than its specific design features, mostly resulting in algorithms that are secure up to a certain computational complexity defined by their key length, number of rounds, and design parameters. Generic attacks can also be devised using linear cryptanalysis, a powerful cryptanalytic method introduced by Matsui [Mat94a] in the early 1990s. A significant enhancement to linear attacks is the simultaneous use of multiple linear approximations, known as multiple linear cryptanalysis. Kaliski and Robshaw [KR94] demonstrated that leveraging several linear approximations can reduce the data complexity

of linear cryptanalysis. Subsequently, Hermelin et al. [HCN08, HCN09, HCN19] proposed multidimensional linear cryptanalysis, a variant of multiple linear cryptanalysis where the masks form a vector space.

The Feistel structure is a well-known and widely analyzed design pattern for block ciphers. These ciphers are known for their efficiency and for balancing security and performance, making them suitable for a wide range of cryptographic applications. One of the primary focuses of research on this structure has been the investigation of Generalized Feistel ciphers (GFCs) with more than two branches, specifically the Type 1, Type 2, and Type 3 Feistel networks described by Zheng, Matsumoto, and Imai [ZMI90]. Type 2 and Type 3 GFC can be expressed using Type 1, as shown in Appendix A, verifying that an attack on one type also affects the others. Additionally, other GFC are unbalanced contracting and expanding Feistel constructions defined by Schneier and Kelsey [SK96].

Research on GFC has progressed from Nyberg’s initial study of Type 2 GFC with fixed permutations [Nyb96] to the development of optimized and flexible permutation strategies. Early investigations primarily examined the behavior of the branch diffusion layer and assessed how different permutation strategies affect the security of the corresponding cipher. Within this context, even-odd branch permutations—mappings that send even-indexed input branches to odd-indexed output branches and vice versa—have been extensively studied for Type-2 GFC. Numerous works [BT19, CGT19, DFLM19, Nyb96, SI22, SM10] have demonstrated that suitably chosen even-odd permutations can significantly reduce the number of rounds required to achieve full diffusion while retaining strong resistance against both differential and linear cryptanalysis. These findings have had a notable impact on the development of lightweight and high-performance block ciphers, influencing designs such as *Simpira* [GM16], *Lilliput* [BFMT16], *TWINE* [SMMK13], and *Piccolo* [SIH⁺11]. Subsequent works extended optimal permutation design to larger block sizes [CGT19, DFLM19] and explored alternating permutations for improved diffusion [BT19, SI22]. More recently, Shibutani and Iwata [SI22] showed that alternating between two distinct branch permutations across rounds can substantially enhance diffusion in 10- and 12-branch Type-2 GFC _{π} constructions. Recent studies also analyzed non-even-odd permutations and showed that at least one even-odd permutation remains optimal up to 32 branches [DDGP22]. Similarly, in [YI11], the authors focus on improving the diffusion layer of Type 1, unbalanced expanding and contracting Feistel constructions, and provide a list of optimal permutations.

Numerous studies in the literature [BL22, GJNS17, HR10, Jut98, NR99, Pat10, SK96, SGW20, YPL11] have focused on the maximum number of rounds that can be distinguished from the random distribution for various t -branch GFC. While some works focus on birthday-bound attacks, others employ different attack models, such as chosen-ciphertext attacks. In contrast, our generic attacks are based on the known-plaintext attack (KPA) model, allowing us to directly compare our results with existing analyses, including the KPA model [NVP13, PNB06, PNB07, THW17]. In [NVP13, THW17], authors use differential cryptanalysis to provide a generic attack on GFC, such as Type 1 and Type 2. Similarly, authors summarize generic attack results for unbalanced contracting GFC in [PNB06] and [PNB07]. Due to the lack of a generic key recovery attack in GFC, our comparisons are limited to unbalanced contracting GFC with the results presented in [BL22] and [GJNS17], which are based on truncated differential attacks and meet-in-the-middle attack, respectively. On the other hand, prior work has explored the security of GFC under different branch permutations, with improvements in diffusion and differential resistance demonstrated through optimal permutation strategies and the use of multiple MDS matrices [CGT19, DDGP22, DFLM19, SV18, SM10, YI11]. Attacks on lightweight ciphers like *SKINNY*, *TWINE*, and *LBlock* [SSD⁺18] revealed that some design permutations offer resistance. While these structural advances have improved diffusion and resistance to differential attacks, the effect of branch permutations on multidimensional linear cryptanalysis remains an open question.

Contribution This paper presents a comprehensive framework for generic attacks on GFC¹, utilizing non-invertible round functions. As discussed in [OBR23], this design feature introduces a specific vulnerability via multidimensional linear cryptanalysis in Feistel ciphers. Employing key-independent linear trails enables computing the distribution of plaintext-ciphertext pairs, thereby facilitating the likelihood-ratio (LLR) test as the optimal test statistic for GFC. Building on [OBR23], we also develop generic key-recovery attacks that combine multidimensional distinguishers with the Fast Fourier Transform (FFT), resulting in significant improvements in data and time complexities for Type 1, Type 2, and Unbalanced contracting GFC (U-Type 1).

In the literature, many studies examine how different permutation strategies affect GFC security, leading to the development of optimized, flexible permutation strategies. We also analyze the diffusion layer of GFC under multidimensional linear attack. One of the main contributions of this work is that using different branch permutations has no negative impact on the applicability or complexity of the multidimensional linear attacks. Our theoretical proofs of Proposition 1, 2, 3, and 4 confirm that changing branch permutations does not change the maximum number of rounds for generic attacks and corresponding attack complexities. In particular, no branch permutation increases the complexity of the attacks; indeed, if a non-even-odd permutation is used for Type 2, it is possible to reduce the data complexity of an r -round attack where r is not a multiple of the branch number. A further contribution of this work is that we significantly extend the best-known bounds on the maximum number of rounds for generic attacks on GFC with arbitrary branch permutation. For distinguishing attack, we reach $t^2 + 2t - 1$ rounds for Type 1 GFC, surpassing the results in [NVP13]. For Type 2 GFC, we enhance the best-known result [NVP13], reaching $2t + 3$ rounds. For U-Type 1, we achieve $t^2 + 2t - 1$ rounds, outperforming the work in [PNB06]. Beyond increasing the maximum number of rounds for distinguishers, our results also significantly reduce the data complexities of reduced-round algorithms, as seen in Table 1. Similarly, we present max number of rounds for key recovery attack on (U)-Type 1 and Type 2, where our approach achieves $t^2 + 2t - 2$ and $3t$ rounds, respectively, for $k = 2^{tn}$ and $t^2 + 3t - 2$ and $4t$ rounds, respectively, for $k = 2^{2tn}$, surpassing previous works as seen in Table 2. To the best of our knowledge, this work is the first generic attack on GFC with arbitrary branch permutation.

Table 1: Theoretical data-complexity q for the r -round distinguishers of GFC with 4 branches. All work efforts and data-complexities are given up to constant factors. All distinguishers are in the known-plaintext model.

Type 1			Type 2		
r	q	Ref.	r	q	Ref.
4	1	§ 3	2	1	§ 3
	$2^{n/2}$	[THW17] [NVP13]		$2^{n/2}$	[NVP13]
5-7	1	§ 3	3	1	§ 3
	2^n	[THW17] [NVP13]		2^n	[NVP13]
8	2^n	§ 3, [THW17, NVP13]	5	2^n	§ 3
9-11	2^{2n}	§ 3	7	2^{2n}	§ 3
	2^{2n}	[NVP13, THW17]		$2^{5n/2}$	[NVP13]
12	2^{2n}	§ 3, [THW17, NVP13]	8	2^{3n}	§ 3, [NVP13]
13-15	2^{2n}	§ 3	9	2^{3n}	§ 3
	2^{3n}	[THW17]		$2^{7n/2}$	[NVP13]
	$2^{5n/2}$	[NVP13]	10	2^{4n}	§ 3, [NVP13]
16-18	2^{3n}	§ 3, [NVP13]	11	2^{4n}	§ 3
19	2^{3n}	§ 3		2^{4n}	§ 3
	2^{4n}	[NVP13]			
20-22	2^{4n}	§ 3, [NVP13]			
23	2^{4n}	§ 3			

¹Due to the better complexity results, our primary focus in this work is on Type 1, Type 2, and U-Type 1 GFC. Detailed analysis of multidimensional linear attacks on (U)-Type 3 is provided in Appendix C.

Furthermore, experimental validation of our theoretical results is provided through application to the CAST-256, and GMiMC_{crf}[4n, 4, n] ciphers, which are given in Figure 6, Table 5, and Table 6 (for 4 and 8 rounds only). Also, we demonstrate improved data complexity for distinguishing and key-recovery attacks in many of the cases presented in Section 6. Specifically, for CAST-256, we achieve 18- and 24-round key-recovery attacks with data complexities of 2^{64} and 2^{96} , respectively, outperforming the results in [WWH09]. Additionally, a 25-round key-recovery attack is feasible with a data complexity of 2^{96} and a time complexity of 2^{222} . For GMiMC_{crf}[4n, 4, n], we improve upon the data and time complexities reported in [GJNS17]. Our 16-round key recovery attack achieves both data and time complexities of 2^{64} , improving upon the respective complexities of 2^{112} and 2^{96} in [GJNS17]. Moreover, our results show that a 22-round attack on GMiMC_{crf}[4n, 4, n] with a data complexity of 2^{96} and a time complexity of 2^{101} is possible. We could not obtain the maximum number of rounds for distinguishing and key-recovery attacks on CAST-256, likely because a single trail performs better at higher rounds. Nevertheless, our approach significantly improves the data and/or time complexities for attacks on the reduced-round version of CAST-256 as compared to prior works [WWH09, ZWW14], and for GMiMC_{crf}[4n, 4, n] over $\mathbb{F}_2^n$ (with even n), compared to existing results in [BL22, GJNS17]. Overall, we have the maximum number of rounds for generic attacks and/or enhanced data complexities and/or improved time complexities in many interesting cases for Type 1, Type 2, and U-Type 1 GFC with an arbitrary branch permutation, making our method more robust.

Table 2: Maximum number of rounds for key recovery attack r_k on 4 branches U-Type 1 GFC with block size $4n$ and key length k . All work efforts and data complexities are given up to constant factors.

k	r	Data	Ref.	Work effort
4n	16	2^{2n}	§ 5	2^{2n}
		$2^{7n/2}$	[GJNS17]	2^{3n}
	17	2^{2n}	§ 5	2^{2n}
		2^{2n}	[BL22]	2^{3n}
8n	22	2^{3n}	§ 5	$n2^{3n}$
	24	2^{3n}	§ 5	$n2^{5n}$
		2^{7n}	[GJNS17]	2^{7n}
	26	2^{3n}	§ 5	$n2^{7n}$

2 Preliminaries

Before exploring the main topic, we present fundamental concepts and definitions that will be used throughout this paper.

2.1 Linear cryptanalysis

The correlation matrix of the function $F_i : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$, denoted by C^{F_i} [DGV95], is a $2^n \times 2^n$ matrix with coordinates

$$C_{v,u}^{F_i} = \mathbb{C}(v^T F_i(x) + u^T x) = \frac{1}{2^n} \sum_{x \in \mathbb{F}_2^n} (-1)^{v^T F_i(x) + u^T x} \quad (1)$$

for each input mask $u \in \mathbb{F}_2^n$ and output mask $v \in \mathbb{F}_2^n$.

In linear cryptanalysis, the objective is to identify linear approximations with high absolute correlations. For iterative functions $F = F_r \circ F_{r-1} \circ \dots \circ F_1$, a tuple of $r+1$ masks

$(\alpha_1, \alpha_2, \dots, \alpha_{r+1})$ defines a linear trail, where α_i and α_{i+1} are input and output masks of the i^{th} round function. The correlation of such a trail is calculated by the product of the correlations of the individual functions F_i . Then, the correlation of a linear approximation is the sum of the correlations of all linear trails that share the same input and output masks as the approximation. Thus, the piling-up lemma [Mat94b] applies when one trail dominates, implying the correlation is approximately that of the dominant trail. For key-alternating ciphers, the round functions are parameterized by round keys such that $F_i(x) = G(x) \oplus k_i$. Consequently,

$$C_{\alpha_{r+1}, \alpha_1}^F = \sum_{\alpha_2, \dots, \alpha_r} \prod_{i=1}^r (-1)^{\alpha_{i+1}^T k_i} C_{\alpha_{i+1}, \alpha_i}^G. \quad (2)$$

The overall absolute correlation of the approximation depends on the key, and Equation (2) indicates that the keys affect only the signs of the trail correlations. Multidimensional linear cryptanalysis examines multiple linear approximations where the respective masks form a vector space. Multidimensional linear approximations are crucial for distinguishers based on the probability distribution of linear projections of plaintext and ciphertext. The key theoretical result is Theorem 1, known as the Poisson summation formula in Fourier analysis. In the context of cryptanalysis, the underlying idea is originally from [BJV04] and is further discussed in [HCN08]. However, the formulation given in Theorem 1 differs in that it applies to an arbitrary vector space of masks. Consequently, using this version requires a change of basis to express the problem in the general form needed for multidimensional attacks.

Theorem 1 (Theorem 4.1. [Bey21], Theorem 1 [OBR23]). *Let $\mathbf{z}$ be a random variable on $\mathbb{F}_2^d$, $V \subseteq \mathbb{F}_2^d$ a vector space, and $V^\perp$ the orthogonal complement of the subspace V . For any $\eta \in \mathbb{F}_2^d$ it holds that*

$$p(\eta) = \Pr(\mathbf{z} = \eta \bmod V^\perp) = \frac{1}{|V|} \sum_{v \in V} (-1)^{v^T \eta} C(v^T \mathbf{z})$$

where $C(v^T \mathbf{z}) = 2 \Pr[v^T \mathbf{z} = 0] - 1$ is the correlation of $v^T \mathbf{z}$.

The theorem in [BJV04] applies when the space contains all possible values of a subset of bits. Our setting is more general as we consider general vector spaces. The special nature of multidimensional linear approximations is highlighted in Theorem 1: when V forms a vector space, the covariance matrix of the estimated correlations is completely determined by the correlations of the linear approximations within V . The underlying reason is that a multidimensional linear approximation can be interpreted as a linear projection of plaintext–ciphertext pairs. Consequently, the statistical dependencies captured by these approximations arise directly from the algebraic relationships induced by this projection. Additional discussion and a more detailed justification of this interpretation are provided in Section 2.2.

2.2 Multidimensional linear distinguisher

A cryptanalyst’s primary objective is to construct a distinguisher or recover the secret key. In linear cryptanalysis, a distinguisher estimates the correlation of a linear approximation using q pairs of plaintext–ciphertext. Considering a function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$, if the correlations $C_{v,u}^F$ for approximations $(u, v) \in \Lambda$ are known, including their signs, the probability distribution of the linear projections of the plaintext and ciphertext can be fully determined. For multidimensional linear cryptanalysis, Theorem 1 can be applied to the vector space $V = \Lambda$ and the random variable $\mathbf{z} = (\mathbf{x}, F(\mathbf{x}))$, where $\mathbf{x}$ is uniform random on $\mathbb{F}_2^n$. This

gives the following result [BR25]:

$$\Pr((\mathbf{x}, F(\mathbf{x})) = (t_1, t_2) \bmod \Lambda^\perp) = \frac{1}{|\Lambda|} \sum_{(u,v) \in \Lambda} (-1)^{u^T t_1 + v^T t_2} C_{v,u}^F. \quad (3)$$

where $\Lambda^\perp = \{(\mathbf{x}, F(\mathbf{x})) \in \mathbb{F}_2^n \times \mathbb{F}_2^n \mid u^T \mathbf{x} + v^T F(\mathbf{x}) = 0 \text{ for all } (u, v) \in \Lambda\}$. In other words, the correlations $C_{v,u}^F$ with $(u, v) \in \Lambda$ completely determine the probability distribution of the linear projections of plaintext and ciphertext, which is a linear map from $\mathbb{F}_2^n \times \mathbb{F}_2^n$ to $\mathbb{F}_2^n \times \mathbb{F}_2^n / \Lambda^\perp$. The quotient space $(\mathbb{F}_2^n \times \mathbb{F}_2^n) / \Lambda^\perp$ is a vector space of the same dimension as Λ . In light of Equation (3), instead of estimating the correlations of linear approximations in Λ , one can estimate the probability distribution of $\pi_{\Lambda^\perp}(\mathbf{x}, F(\mathbf{x}))$ for uniform random $\mathbf{x}$.

Adversaries aim to maximize the advantage defined as $\text{Adv} = P_S - P_F$ where P_S (success probability) is the rate of correctly identified positive instances and P_F (false positive rate) is the rate of negative cases incorrectly identified as positive. The data complexity q required for a distinguishing attack to achieve specific values of P_S and P_F depends on the statistical test used. For both known and unknown correlation sign cases, the data complexities are inversely proportional to the capacity of the multidimensional linear approximation Λ over the function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$, which is defined as $\text{Cap}(\Lambda) = \sum_{(u,v) \in \Lambda} (C_{v,u}^F)^2$. When correlation signs are unknown, precise probability distribution estimation involves guessing, and Vaudenay [Vau96] proposed using the χ^2 test to detect non-uniformity, resulting in a data complexity of $q \approx \sqrt{|\Lambda|} / \text{Cap}(\Lambda)$ [Nyb13]. On the other hand, if correlations are known, the likelihood-ratio test proposed by Vaudenay et al. [BJV04] can be used, with a data complexity of $q \approx 1 / \text{Cap}(\Lambda)$ [Nyb13]. Thus, the data complexity for known signs is significantly lower, by a factor of the square root of the number of approximations, compared to the unknown signs case. Further details on the asymptotic data complexities of the distinguishers are provided in Appendix B.

In standard linear cryptanalysis, the data complexity is inversely proportional to the square of the correlation of a single approximation. Consequently, the known-sign case in multidimensional linear cryptanalysis significantly reduces data complexity compared to the unknown-sign case and to standard linear cryptanalysis.

2.3 Key Recovery Attack

A distinguisher is typically only the first stage in mounting a full attack on a cipher. It can be extended into a key-recovery attack by appending/prepending one or more rounds to the distinguisher and guessing the round key involved in those additional rounds. The naive approach evaluates and stores the estimated absolute correlation for all 2^ℓ round-key guesses, giving a total time complexity of $\mathcal{O}(q2^\ell)$ for q plaintext-ciphertext pairs. Matsui [Mat94b] proposed an improved technique based on the observation that the linear approximation can be decomposed into two independent parts—one dependent on the data and one on the round key. Indeed, the estimated correlation for any key guess k can be expressed as

$$\hat{c}_k = \frac{1}{q} \sum_{i=1}^q (-1)^{u^\top x_i + v^\top G_k^{-1}(y_i)} = \frac{1}{q} \sum_{z \in \mathbb{F}_2^\ell} (-1)^{v^\top G_k^{-1}(z)} \sum_{i=1}^q (-1)^{u^\top x_i} \delta_{z, y_i}$$

where G_k corresponds to the appended or prepended rounds. Matsui's method reduces the time complexity to $\mathcal{O}(q + 2^{2\ell})$ by decomposing the approximation into two parts and precomputing a $2^\ell \times 2^\ell$ table. Separating the expression into a matrix-vector product precisely enables an efficient extension, which motivates defining a matrix M with coordinates indexed by $(k, z) \in \mathbb{F}_2^\ell \times \mathbb{F}_2^\ell$ and a vector w by

$$M_{k,z} = (-1)^{v^\top G_k^{-1}(z)} \quad \text{and} \quad w_z = \frac{1}{q} \sum_{i=1}^q (-1)^{u^\top x_i} \delta_{z, y_i}, \quad (4)$$

so that the vector of estimated correlations is calculated as $\hat{c}_k = Mw$. The FFT-based improvement of Collard et al. [CSQ07] exploits that, when the key guess k is XORed directly to the plaintext/ciphertext, the matrix M is circulant, allowing Mw to be computed using the FFT in $\ell 2^\ell$ arithmetic operations. As a result, the total time complexity is reduced to $\mathcal{O}(q + \ell 2^\ell)$, a significant improvement over Matsui's original approach.

In multidimensional linear cryptanalysis, the same computation must be performed for every approximation. The notion of separability defined in Equation (4) is therefore crucial, as without it the entire expression must be recomputed for every pair of masks u and v . With the separation, the matrix M depends on the mask v , while the vector w depends on the mask u .

2.4 Generalized Feistel Ciphers

Generalized Feistel Ciphers (GFC) are block cipher structures that generalize the classic Feistel network by splitting the input into multiple branches. The structure of the round transformation depends on the types of GFC. Type 1 GFC applies a round function to only the first branch, whose output is XORed with the second branch. U-Type 1 GFC uses a contracting function on all but the first branch to update the first. Type 2 GFC modifies the odd-indexed branches, while the others remain unchanged in each round. Then the branch permutation is applied to all branches, which is a deterministic mapping over t blocks, typically as a left cyclic shift, which is $\pi_s(X_0, X_1, \dots, X_{t-1}) = (X_{t-1}, X_0, X_1, X_2, \dots, X_{t-2})$, for standard GFC as illustrated in Figure 1.

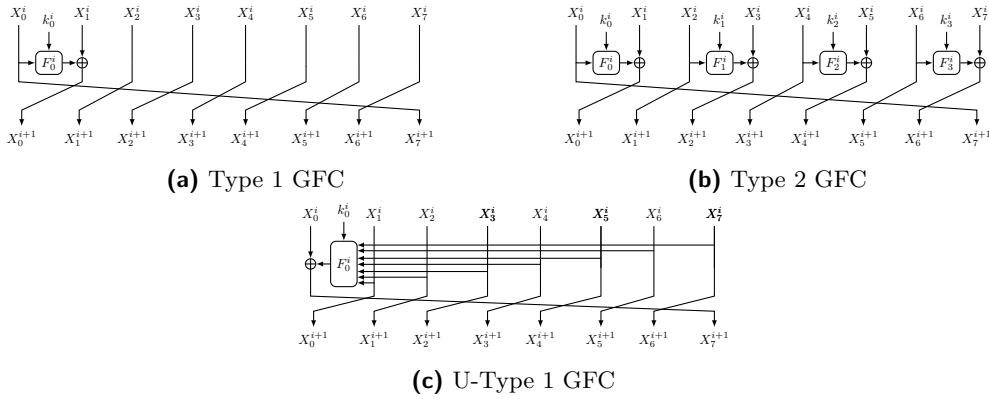


Figure 1: Round transformation of GFC with 8 branches.

2.5 CAST-256

CAST-256 [AG99] is a symmetric-key block cipher candidate for the Advanced Encryption Standard, designed as an extended version of CAST-128. It is a 4-branch GFC that operates on 128-bit plaintext blocks and supports key lengths ranging from 128 to 256 bits. Specifically, CAST-256 has 48 rounds, with a 256-bit key. Each round in CAST-256 involves a Feistel structure with three distinct round functions F_1 , F_2 , F_3 , identical in structure but differing in the operations used: addition, subtraction, and XOR. Each round function starts with key addition and key-dependent rotation. Although not selected as the final AES standard, CAST-256 remains widely analyzed as a block cipher.

2.6 GMiMC

GMiMC [AGP⁺19] (Generalized MiMC [AGR⁺16]) is a block cipher designed to optimize cryptographic operations, particularly for multi-party computation and zero-knowledge proof systems. It employs a generalized Feistel structure over $\mathbb{F}_p$ where p is a prime number or $\mathbb{F}_{2^n}$. GMiMC_{crf} and GMiMC_{erf} are two variants of the GMiMC block cipher. The round function $F(x)$ of GMiMC_{crf}, and GMiMC_{erf} typically involves a non-linear monomial x^3 . In $\mathbb{F}_{2^n}$, the monomial x^3 is a permutation if and only if $\gcd(3, 2^n - 1) = 1$, which holds when n is odd but not when n is even. By choosing n to be even, x^3 becomes a non-permutation, introducing non-invertibility in the round function $F(x)$. So, we analyze GMiMC over $\mathbb{F}_{2^n}$ where n is even in this work. GMiMC_{crf} employs a contracting round function in a generalized Feistel structure (U-Type 1) and operates on the XORed inputs of $t - 1$ branches along with the round key and constant. The contracting nature reduces the size of the output relative to the inputs. In contrast, GMiMC_{erf} utilizes an expanding round function (U-Type 3), producing a larger output from smaller inputs. GMiMC_{erf} operates on the XORed input of one branch, the round key, and the constant. The main parameters of block ciphers are represented as $[k, t, N]$. For instance, GMiMC_{crf}[$4N, 4, N$] specifies the GMiMC with a contracting round function (CRF), where N is the branch size, $4N$ is the key size, and the number of branches is 4.

3 Generic Multidimensional Linear Distinguishers

In the context of multidimensional linear cryptanalysis of GFC, the primary objective is to minimize data complexity by keeping the overall correlation of the trail as small as possible, which is achieved by skipping as many rounds as possible without contributing to the correlation. Two fundamental operations in the GFC -forking and XOR- strictly constrain how linear masks propagate through the cipher and the resulting correlation. To keep the correlation constant, the masks of all branches must be identical for the XOR operation. In contrast, the forking operation enforces that the XOR of two branch masks equals the mask of the remaining branch. If a non-zero mask propagates through the XOR operation, the corresponding round function is activated, contributing a non-zero correlation to the trail. Hence, by carefully positioning the non-zero mask, an attacker can effectively reduce the number of active round functions, thereby decreasing the data complexity of the attack.

Our attacks target GFC with non-invertible functions similar to those in [OBR23]. The analysis here is generic, assuming the public functions $F_1, F_2, \dots, F_r$ are chosen independently and randomly. The signs of the correlations of linear trails in a key-alternating cipher with independent round keys are shown to be key-dependent. On the other hand, key-independent trails exist for GFC, even with independent round keys. These key-independent trails can significantly reduce the data complexity of multidimensional linear attacks. To determine the correlation of the iterative trail, we use the following result by Daemen and Rijmen. The formulation in Theorem 2 is due to [Bey21].

Theorem 2 (Theorem 3.1. [Bey21]). *Let $\mathbf{c}$ be the correlation of a nontrivial linear approximation for a uniform random function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$. The random variable $2^{n-1}(\mathbf{c}+1)$ follows a binomial distribution with mean 2^{n-1} and variance 2^{n-2} . In particular, as n approaches infinity, the distribution of $2^{n/2}\mathbf{c}$ converges to the standard normal distribution $\mathcal{N}(0, 1)$.*

3.1 Data-complexities of generic multidimensional linear distinguisher of r -round and t -branch Type 1 and U-Type 1 GFC

We begin by describing the mask spaces used in multidimensional linear distinguishers for Type 1 and U-Type 1 GFC.

Defining the mask space A multidimensional linear distinguisher operates on a specific vector space of input-output mask pairs. For 4 (or multiples of 4) rounds (U)-Type 1, the vector space of input-output mask pairs is equal to

$$\Lambda_{T_1} = \left\{ ((u, 0, \dots, 0, 0), (u, 0, \dots, 0, 0)) \mid u \in \mathbb{F}_2^n \right\} \text{ for Type 1,}$$

$$\Lambda_{U_1} = \left\{ ((0, 0, \dots, 0, u), (0, 0, \dots, 0, u)) \mid u \in \mathbb{F}_2^n \right\} \text{ for U-Type 1.}$$

as seen in Figure 2. Each nonzero $u \in \mathbb{F}_2^n$ yields one linear approximation, so the distinguisher simultaneously handles $2^n - 1$ approximations. The data complexity of the multidimensional distinguisher is determined by the sum of squared correlations over all linear approximations.

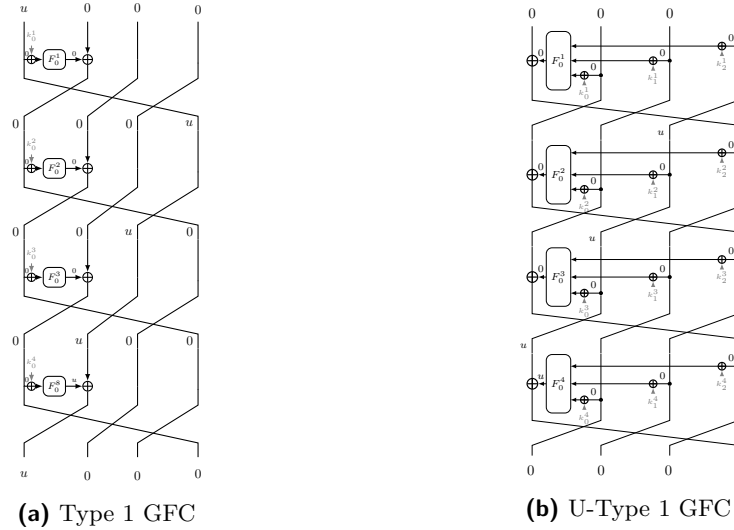


Figure 2: Four-round linear trail for 4 branches (U)-Type 1 GFC with key-independent correlation.

Calculating the capacity Due to the t -branch cyclic permutation, a nonzero mask placed does not influence the next $t - 1$ rounds. This gives an iterative pattern: every t rounds, one round function becomes active. Let $\mathbf{c}_i$ be the correlation of the linear approximation with input mask zero and output mask u over the (uniformly random) function F_{2i-1} . Therefore, the trail correlation is $\mathbf{c}_u = \prod_{i=1}^{\lfloor r/t \rfloor} \mathbf{c}_i$ where r/t is rounded down since we can skip the first $t - 1$ rounds, fixing the nonzero mask in the first branch for Type 1 and in the last branch for U-Type 1.

By Theorem 2, $\mathbf{c}_i$ is a random variable with an asymptotically normal distribution centered around zero and with standard deviation $2^{-n/2}$. Since the random variables are independent by the strong assumption that the random functions $F_1, \dots, F_r$ are independent, the variance of $\mathbf{c}_u$ is equal to $\mathbb{E}(\mathbf{c}_u^2) = 2^{-n \lfloor r/t \rfloor}$. The trail in Figure 2 is dominant since one round function is active in t rounds. Thus, by the piling-up lemma, the

sum of the squared correlations is well approximated by the sum of the squares of the trail correlations, which can be calculated in time $\mathcal{O}(n2^n)$ given the round functions $F_1, \dots, F_r$. A multidimensional linear distinguisher can be set up by simultaneously working with all possible masks u . In the generic case, the sum of squared correlation is a random variable. However, since its distribution is strongly concentrated around the mean, its average is a good estimate:

$$\mathbb{E} \left(\sum_{u \in \mathbb{F}_2^n \setminus \{0\}} \mathbf{c}_u^2 \right) \approx 2^n \mathbb{E}(\mathbf{c}_u^2) = 2^{n-n\lfloor r/t \rfloor}.$$

Resulting data complexity As discussed in Section 2.2, the central question is whether the sign of the correlation can be determined, leading to the use of a specific test statistic. In other words, if it is possible to remove the key from linear approximation and obtain a key-independent trail, the sign of the correlation is known by Equation (2) and the LLR method can be used as the optimal test statistic. As illustrated by the linear trail in Figure 2, all input masks are zero, ensuring that such key-independent trails indeed exist for Type 1 and U-Type 1 GFC. For the LLR distinguisher, the required amount of data is inversely proportional to this sum of squared correlations. Thus, the asymptotic data complexity becomes $q = \mathcal{O}(2^{n\lfloor r/t \rfloor - n})$ for Type 1 and U-Type 1 GFC.

3.2 Data-complexities of generic multidimensional linear distinguisher of Type 2 GFC

The data complexity is computed in the same way as given in Section 3.1. A multidimensional linear distinguisher can be established by using all possible masks u simultaneously, which can be constructed as $\Lambda = \{((u, 0, \dots, 0), (u, 0, \dots, 0)) \mid u \in \mathbb{F}_2^n\}$ for Type 2. As observed in Figure 3, the linear trail activates only one round function F_{2i} every two rounds for Type 2 GFC. Moreover, the first round remains inactive, fixing a non-zero mask in an even-indexed block. Consequently, for $r \geq 2$ rounds, its correlation is given by $\mathbf{c}_u = \prod_{i=1}^{\lfloor r/2 \rfloor} \mathbf{c}_i$. Under the strong assumption that the random functions $F_1, \dots, F_r$ are independent, the associated random variables are also independent. Consequently, the variance of $\mathbf{c}_u$ is given by $\mathbb{E}(\mathbf{c}_u^2) = 2^{-n\lfloor r/2 \rfloor}$. The next step is to compute the capacity. Let $\mathbf{c}_i$ denote the correlation of the linear approximation with an input mask of zero and an output mask u over the uniformly random function F_i where $i \in \{1, 2, \dots, r/2\}$. According to Theorem 2, $\mathbf{c}_i$ is a random variable that follows an asymptotically normal distribution with a mean of zero and a standard deviation of $2^{-n/2}$. Then we get

$$\mathbb{E} \left(\sum_{u \in \mathbb{F}_2^n \setminus \{0\}} \mathbf{c}_u^2 \right) \approx 2^{n-n\lfloor r/2 \rfloor}.$$

As shown in Figure 3, all input masks are non-zero, so key-independent trails exist for Type 2 GFC, leading to the use of the LLR method. Thus, the data-complexity of is $q = \mathcal{O}(2^{n\lfloor r/2 \rfloor - n})$.

Note that the nonzero mask position should be placed at an even index. After adjusting the input mask position, the output mask position is determined according to the structure and the number of branches. For example, the vector space for the 4-round linear trail in Figure 3 is $\Lambda = \{((u, 0, 0, 0, 0, 0, 0, 0), (0, 0, 0, 0, u, 0, 0, 0)) \mid u \in \mathbb{F}_2^n\}$. Furthermore, the vector space Λ used to achieve the data complexity in Section 3.2 is not unique. In the context of Type 2 GFC, valid linear trails can be constructed by activating a single round function every two rounds through the assignment of a nonzero mask at an even index. An alternative one is defined as $\Lambda = \{((0, 0, \dots, u, 0), (0, 0, \dots, u, 0)) \mid u \in \mathbb{F}_2^n\}$. More generally, for a Type 2 GFC with t branches, there exist $t/2$ such even indices, each giving rise to a distinct vector space that yields the same data complexity.

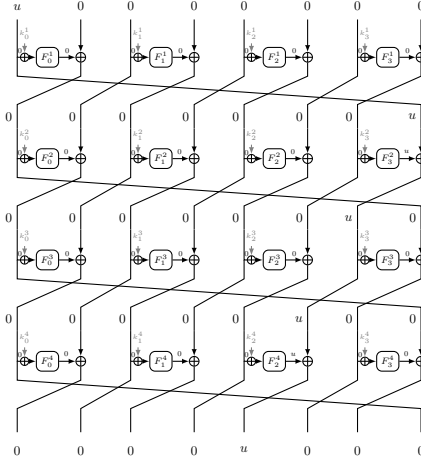


Figure 3: Four-round linear trail for 8 branches Type 2 GFC with key-independent correlation.

3.3 Distinguisher using the likelihood-ratio test

The likelihood ratio test is the most robust statistical method for distinguishing between two hypotheses. When trail correlations are key-independent, the plaintext–ciphertext sum distribution can be precisely determined (cf. Theorem 1), enabling the use of the LLR method. Given samples $z_1, \dots, z_q$, the LLR statistic is

$$\lambda_{\text{real}} = \log \prod_{i=1}^q \frac{p_{\text{real}}(z_i)}{p_{\text{ideal}}(z_i)} = \sum_{i=1}^q \log \frac{p_{\text{real}}(z_i)}{p_{\text{ideal}}(z_i)},$$

where p_{real} denotes the distribution induced by the cipher and p_{ideal} the uniform distribution on $\mathbb{F}_{2^n}$. Let $(x_0^i, \dots, x_t^i)$ and $(y_0^i, \dots, y_t^i)$ denote q plaintext–ciphertext pairs. We define $z_i = x_0^i + y_0^i$ for Type1 and Type2, and $z_i = x_{t-1}^i + y_{t-1}^i$ for U-Type1. As established in Section 2.2, the asymptotic data complexity of the LLR test is inversely proportional to the sum of squared correlations, yielding

$$\begin{aligned} q &= \mathcal{O}(2^{\lfloor r/t \rfloor n - n}) \text{ for Type 1 and U-Type 1 where } r \geq t, \\ q &= \mathcal{O}(2^{\lfloor r/2 \rfloor n - n}) \text{ for Type 2 where } r \geq 2. \end{aligned} \quad (5)$$

Accurate estimates of data complexity are crucial for the practical implementation of attacks. Following the framework of Baignères, Junod, and Vaudenay [BJV04], the LLR statistic λ asymptotically follows a normal distribution, when the sample size increases: $\lambda_{\text{real}} \sim \mathcal{N}(\mu_{\text{real}}, \sigma^2)$, and $\lambda_{\text{ideal}} \sim \mathcal{N}(\mu_{\text{ideal}}, \sigma^2)$. For a fixed false-positive rate P_F , the decision threshold is

$$\tau = -\mu_{\text{real}} + \sqrt{2\mu_{\text{real}}} \Phi^{-1}(1 - P_F), \quad (6)$$

leading to success probability $P_S = \Pr[\lambda_{\text{real}} \geq \tau]$. In Section 6, experimental results are presented to determine the success probability for specified false-positive rates and corresponding data complexities

3.4 Calculating the maximum number of rounds for distinguishing attacks

The maximum number of rounds r_d for the LLR multidimensional distinguisher can be calculated from its data complexity. In the literature, several studies [NVP13, THW17,

PNB06] adopt an upper bound of 2^{tn} for the data complexity. Accordingly, we use this threshold in our distinguishing attack on GFC with t branches and branch size n . Then, for Type 1 and U-Type 1 GFC, we derive the following inequality $2^{n\lfloor r_d/t \rfloor - n} \leq 2^{tn}$. This implies that $\lfloor r_d/t \rfloor \leq t + 1$, and hence $r_d \leq t^2 + 2t - 1$. Similarly, for Type 2 GFC, the condition $2^{n\lfloor r_d/2 \rfloor - n} \leq 2^{tn}$ leads to the conclusion that $r_d \leq 2t + 3$.

4 Diffusion Behavior of Generalized Feistel Ciphers

One important characteristic of GFC is its non-trivial branch permutation, which ensures efficient diffusion across rounds. Hence, the choice of the branch permutation is a critical factor in determining the minimum number of rounds required for full diffusion. In GFC, branches are indexed sequentially from left to right as $0, 1, \dots, t - 1$. Let π be a branch permutation over the set $\{0, 1, \dots, t - 1\}$, and denote by $\pi(i)$ the index of the branch obtained by applying π to the i -th branch. For instance, the branch permutation is given by $\pi = (3, 0, 1, 2)$, meaning that $\pi(0) = 3$, $\pi(1) = 0$, $\pi(2) = 1$, and $\pi(3) = 2$. For any integer $r \geq 1$, we define $\pi^r(i)$ as the branch permutation that consists of applying π r times. Similarly, $\pi^{-r}(i)$ denotes the result of applying the inverse branch permutation π^{-1} r times.

4.1 Diffusion behavior of Type 1 GFC $_{\pi}$ under multidimensional linear cryptanalysis

In Type 1 GFC $_{\pi}$, a nonlinear round function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is applied to the first (or leftmost) block located at X_0 , then the output is XORed with the next block located at X_1 . Let's call the result of this process X'_1 and X_i where $i \in \{0, 2, 3, \dots, t - 1\}$ will be the same before a branch permutation, so a permutation is applied to $(X_0, X'_1, X_2, \dots, X_{t-2}, X_{t-1})$. Then, we can define one round transformation of Type 1 GFC $_{\pi}$ as follows

$$(X_0, X_1, \dots, X_{t-1}) \rightarrow \pi(X_0, X'_1, \dots, X_{t-2}, X_{t-1}) \rightarrow (X_{\pi[0]}, X'_{\pi[1]}, \dots, X_{\pi[t-2]}, X_{\pi[t-1]}).$$

Consequently, if a non-zero mask is placed at X_1 , the corresponding round function becomes active due to the XOR operation. In contrast, by placing a non-zero mask at a position different from X_0 , it becomes possible to bypass some rounds and reduce the overall data complexity.

In [YI11], authors provide different branch permutations based on branch numbers for Type 1 GFC and evaluate the security of the resulting constructions against various forms of cryptanalysis, including saturation, impossible differential, differential, and linear attacks. Figure 10a illustrates a multidimensional linear cryptanalysis trail for an 8-branch Type 1 GFC $_{\pi}$, where the branch permutation $\pi = (2, 3, 4, 5, 1, 6, 7, 0)$ is given in [YI11]. The trail depicted in Figure 10a exhibits a repetitive structure, allowing it to be extended straightforwardly. By appropriately selecting the position of the non-zero input mask, the first 7 rounds can be bypassed entirely, resulting in only one active round function for every 8 rounds. Note that the data complexity of Type 1 GFC $_{\pi}$ depends on the order of the branch permutation π . In Figure 10a, the branch permutation π forms a single cycle of length t . In the case where π consists of several disjoint cycles, activation occurs for the cycle that includes the non-zero masked index, while all other cycles are omitted from the evaluation. For example, if $\pi = ((0 \ 1 \ 2 \ 3 \ 4)(7 \ 6 \ 5))$ and the non-zero mask is fixed to index 0 in the 5-element cycle, the number of active round functions becomes $\lfloor r/5 \rfloor$ for 8 branches U-Type 1 GFC $_{\pi}$. Notably, branch permutation layers composed of multiple disjoint cycles are a weakness for Type 1 GFC $_{\pi}$, and no such configurations have been identified as optimal. Therefore, we provide the data complexity of t -branch Type 1 GFC $_{\pi}$ in Proposition 1 where π is a single cycle of length t .

Proposition 1. *Let π be a branch permutation of order t over the t branches Type 1 GFC $_{\pi}$, where the round function is $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$. The data complexity of a multidimensional LLR distinguisher spanning r rounds is $q = \mathcal{O}(2^{n\lfloor r/t \rfloor - n})$ for all Type 1 GFC $_{\pi}$.*

Proof. In the Type 1 GFC $_{\pi}$, the XOR operation occurs at the X_1 position. A branch permutation π is a full cycle of length t , $\{\pi^0(i), \pi^1(i), \dots, \pi^{t-1}(i)\}$ contains exactly all t distinct elements in $\{0, 1, \dots, t-1\}$, in some order. This means that over the full range $r = 0$ to $r = t-1$, the value 1 appears exactly once in the sequence with length t . This periodic behavior implies that the trail forms an iterative structure, activating only one round function every t rounds. An example of a one-round linear trail in the Type 1 GFC $_{\pi}$ is

$$(0, 0, \dots, u, \dots, 0) \rightarrow \pi(0, 0, \dots, u, \dots, 0) \rightarrow (0_{\pi[0]}, 0_{\pi[1]}, \dots, u_{\pi[i]}, \dots, 0_{\pi[t-2]}, 0_{\pi[t-1]}),$$

where the non-zero component u is initially placed at index i . To avoid immediate activation of this round function, the non-zero input mask must be assigned to a sub-block index different from X_1 . More precisely, to delay the first activation of round as much as possible, the non-zero input mask should be placed at the index i that maximizes the number of rounds r such that $\pi^r(i) = 1$. Considering $\pi^r(i) \neq 1$ for all $r \in 0, 1, \dots, t-2$, 1 must appear at $r = t-1$, i.e., $\pi^{t-1}(i) = 1$. The number of active rounds is therefore $\lfloor r/t \rfloor$ for the r -round multidimensional distinguisher. The overall trail correlation is given by $\mathbf{c}_u = \prod_{i=1}^{\lfloor r/t \rfloor} \mathbf{c}_i$. Accordingly, the data complexity of the distinguisher is $q = \mathcal{O}(2^{n\lfloor r/t \rfloor - n})$. $\square$

4.2 Diffusion behavior of U-Type 1 GFC $_{\pi}$ under multidimensional linear cryptanalysis

In the U-Type 1 GFC $_{\pi}$, a nonlinear round function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is applied to the XOR of the rightmost $t-1$ sub-blocks. The output of this function is then XORed with the leftmost block, located at position X_0 , yielding a new value denoted by X'_0 . The remaining sub-blocks X_i for $i \in \{1, 2, \dots, t-1\}$ remain unchanged at this stage. A branch permutation π is then applied to the updated block vector. The one-round transformation of the U-Type 1 GFC $_{\pi}$ can thus be described as

$$(X_0, X_1, \dots, X_{t-1}) \xrightarrow{\pi} (X'_{\pi[0]}, X_{\pi[1]}, \dots, X_{\pi[t-1]}).$$

Since the round function output modifies X_0 , placing a non-zero input mask at this position causes the round function to become active. Conversely, placing the mask at any position other than X_0 can potentially delay the activation of the round function, enabling multiple initial rounds to be bypassed and thereby reducing the overall data complexity of attacks.

In [YI11], various optimal branch permutations are provided for U-Type 1 GFC based on the number of branches. Figure 10b illustrates a multidimensional linear cryptanalysis trail for an 8-branch U-Type 1 GFC $_{\pi}$, using the branch permutation $\pi = (1, 2, 3, 4, 5, 6, 7, 0)$ in [YI11]. The trail displays a repetitive structure that allows straightforward extension, following a strategy analogous to that used for Type 1 GFC. By appropriately selecting the input mask position, the first 7 rounds can be bypassed, resulting in 1 active round per 8 rounds. The data complexity of U-Type 1 GFC $_{\pi}$, like its Type 1 counterpart, is governed by the order of the branch permutation π , specifically the cycle containing both the index influenced by the XOR and the index with a nonzero mask. Even when π consists of multiple disjoint cycles, only the one containing the masked index contributes to round activation; others remain inactive and are excluded from analysis. Thus, we consider only the case where π is a single cycle in Proposition 2 to generalize our result for U-Type 1 GFC $_{\pi}$.

Proposition 2. *Let π be a branch permutation of order t over the t branches U-Type 1 GFC $_{\pi}$, where the round function is $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$. The data-complexity of the r rounds multidimensional LLR distinguisher is $q = \mathcal{O}(2^{n\lceil r/t \rceil - n})$ for all U-Type 1 GFC $_{\pi}$.*

Proof. As in the Type 1 GFC, the U-Type 1 GFC $_{\pi}$ uses a single nonlinear round function per round, with the XOR operation taking place at position X_0 . Given that the branch permutation π forms a full cycle, $\{\pi^0(i), \pi^1(i), \dots, \pi^{t-1}(i)\}$ covers all t indices exactly once, in some order. This means that over the full range $r = 0$ to $r = t - 1$, the value 0 appears exactly once in the sequence with length t . This structure naturally leads to an iterative trail pattern, wherein one round function becomes active every t rounds. A one-round linear trail in the U-Type 1 GFC $_{\pi}$ can be expressed as

$$(0, 0, \dots, u, \dots, 0) \xrightarrow{\pi} (0_{\pi[0]}, 0_{\pi[1]}, \dots, u_{\pi[i]}, \dots, 0_{\pi[t-1]}),$$

where the non-zero input mask u is initially placed at index i . Avoiding round function activation in the first round requires choosing $i \neq 0$. The analysis follows a similar way to that in the proof of Proposition 1. Specifically, to delay the first activation of the round function, the non-zero mask should be positioned at an index i that maximizes the number of rounds r for which $\pi^r(i) = 0$. When the index i is chosen such that $\pi^r(i) \neq 0$ for all $r = 0, 1, \dots, t - 2$, then due to the cyclic nature of π , it must hold that $\pi^{t-1}(i) = 0$. This ensures the round function can be activated after the first $t - 1$ rounds. As a result, an r -round multidimensional LLR distinguisher activates $\lceil r/t \rceil$ rounds, and the corresponding data complexity is therefore $q = \mathcal{O}(2^{n\lceil r/t \rceil - n})$. $\square$

4.3 Diffusion behavior of Type 2 GFC $_{\pi}$ under multidimensional linear cryptanalysis

In Type 2 GFC $_{\pi}$, a nonlinear round function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is applied to the blocks located at even indices X_{2i} where $i \in \{1, 2, \dots, (t-2)/2\}$, then the outputs are XORed with the blocks located at odd indices X_{2i+1} . Let's call the result of this process X'_{2i+1} and X_{2i} will be the same before a branch permutation, so a branch permutation is applied to $(X_0, X'_1, \dots, X_{t-2}, X'_{t-1})$. Then, we can define one round transformation of Type 2 GFC $_{\pi}$ as follows

$$(X_0, X_1, \dots, X_{t-1}) \rightarrow \pi(X_0, X'_1, \dots, X_{t-2}, X'_{t-1}) \rightarrow (X_{\pi[0]}, X'_{\pi[1]}, \dots, X_{\pi[t-2]}, X'_{\pi[t-1]}).$$

In Type 2 GFC, if a non-zero mask is placed at an odd index, the corresponding round function becomes active due to the XOR operation. On the other hand, by placing non-zero masks at even indices, it is possible to bypass certain rounds due to the forking operation and thereby reduce the overall data complexity. Even-odd branch permutations are defined as index mappings where even-numbered input blocks correspond to odd-numbered output blocks and vice versa. Several works in the literature [BT19, CGT19, DFLM19, Nyb96, SI22, SM10] investigate enhancing the security of Type 2 GFC by employing various even-odd branch permutations. Specifically, research on optimal even-odd branch permutations aims to minimize the diffusion round while maintaining robustness against differential and linear cryptanalysis. In [SM10], Suzaki and Minematsu present an example of 8-branch Type 2 GFC $_{\pi}$. We analyze this construction under multidimensional linear cryptanalysis, as illustrated in Figure 4a, and show that the resulting data complexity matches that of the standard Type 2 GFC given in Section 3.2 since there is one active round function every two rounds. An alternative research direction explored whether using multiple distinct branch permutations across rounds could enhance diffusion [BT19, SI22]. In particular, Shibutani and Iwata [SI22] explore the use of two different branch permutations to enhance the diffusion in 10- and 12-branch Type 2 GFC $_{\pi}$. However, our analysis shows that even when different even-odd branch permutations are used in each round, the resulting

data complexity under multidimensional linear cryptanalysis remains unchanged. This is because the structural property, where even-indexed inputs are mapped to odd-indexed outputs, ensures that only one round function is activated every two rounds, regardless of branch permutation variation. As illustrated in Figure 4b, this behavior leads to the same data complexity of $q = \mathcal{O}(2^{n\lfloor r/2 \rfloor - n})$. These results are then generalized to all Type 2 GFC_π employing even-odd branch permutations, as formally stated in Proposition 3.

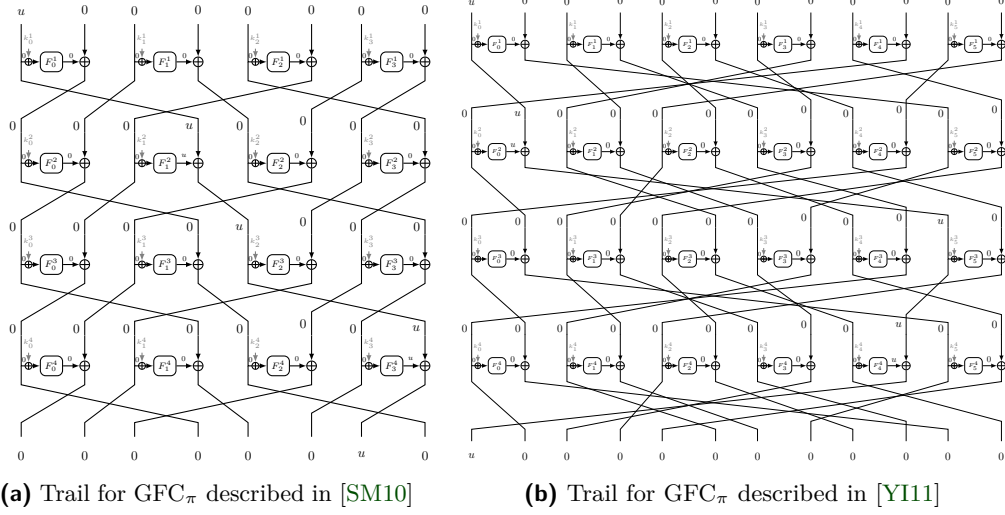


Figure 4: Four-round linear trail for Type 2 GFC_π described by Suzuki and Minematsu [SM10], and by Shingo and Iwata [YI11] with key-independent correlations.

Proposition 3. *Let π be an arbitrary even-odd branch permutation of a t -branch Type 2 GFC_π with round functions $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$. The data-complexity of the r rounds multidimensional LLR distinguisher is defined as $q = \mathcal{O}(2^{n\lfloor r/2 \rfloor - n})$ for all Type 2 GFC_π .*

Proof. Assume a nonzero input mask u is placed at an even index (e.g., position X_0), while all other positions are set to zero. A one-round trial of Type 2 GFC_π can be defined as

$$(u, 0, \dots, 0) \rightarrow \pi(u, 0, \dots, 0, 0) \rightarrow (u_{\pi[0]}, 0_{\pi[1]}, \dots, 0_{\pi[t-2]}, 0_{\pi[t-1]}).$$

Here, the branch permutation π governs the propagation of mask values through the trail. In the case of an even-odd branch permutation, where each even index is mapped to an odd one, the initial mask at an even-indexed position is routed to some X_{2i+1}^2 where $i \in \{0, 1, \dots, (t-2)/2\}$, activating a round function in the second round. In the subsequent round, the output is routed back to an even-indexed position, which does not activate the round function. As a result, round functions are activated only once in every other round, yielding one active round function every two rounds. This behavior leads to the same data complexity as given in Section 3.2.

More generally, the number of active round functions corresponds to the number of odd-indexed positions that carry nonzero masks throughout the trail. The overall correlation c_u of a multidimensional linear trail is given by the product of the correlations of the active round functions. For an r -round trail starting with an even-indexed mask, the number of active rounds is $\lfloor r/2 \rfloor$. Therefore, the trail correlation is $\mathbf{c}_u = \prod_{i=1}^{\lfloor r/2 \rfloor} \mathbf{c}_i$, resulting in data complexity as $q = \mathcal{O}(2^{n\lfloor r/2 \rfloor - n})$. $\square$

As shown in the proof of Proposition 3, the data complexity of the multidimensional attack on Type 2 GFC_π does not depend on the order of the even-odd branch permutation.

Consequently, it is not necessary to analyze attack scenarios involving branch permutations composed of multiple disjoint cycles. Regardless of the branch permutation structure, exactly one round function is activated every two rounds.

Recently, Delaune et al. [DDGP22] extended the study of GFC branch permutations by investigating non-even-odd structures and identified a set of optimal non-even-odd branch permutations. In this case, it is possible to obtain better data complexities by skipping many rounds because of repetitive even indexes through r rounds LLR distinguisher where r is not a multiple of the branch number. Figure 11 presents the multidimensional LLR distinguisher of Type 2 GFC with non-even-odd branch permutation $\pi_1 = (3, 18, 5, 16, 7, 12, 9, 10, 1, 14, 13, 2, 15, 8, 11, 21, 17, 4, 19, 6, 0, 20)$. As seen in Figure 11, by fixing the nonzero mask values at index 20 in the first round, we can skip the first 2 rounds of reduced rounds distinguishing attacks. Thus, we can improve the data complexity of reduced round Type 2 GFC $_{\pi}$ if π is a non-even-odd branch permutation by carefully selecting the nonzero mask value index. We generalize this result for all Type 2 GFC $_{\pi}$ with non-even-odd branch permutation, which is given in Proposition 4.

Proposition 4. *Let π be a branch permutation of order t over the t -branch Type 2 GFC $_{\pi}$ where the round function is $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$. The data-complexity of the r rounds multidimensional LLR distinguisher of Type 2 GFC $_{\pi}$ is defined as $q = \mathcal{O}(2^{n(N_o)-n})$, where N_o is the number of odd indexes that the nonzero mask value propagates through the r rounds trail, where $r < kt$ with $k \in \mathbb{Z}^+$. If $r = kt$, then we have $q = \mathcal{O}(2^{n\lfloor r/2 \rfloor - n})$.*

Proof. While even-odd branch permutations yield predictable diffusion patterns, analyzing non-even-odd branch permutations requires a more refined examination of how the branch permutation π propagates non-zero masks. The behavior can be categorized as follows when

1. an even index maps to an odd index, the corresponding round function is activated.
2. an even index maps to another even index, the round function can be bypassed.
3. an odd index maps to another odd index, the corresponding round function is activated.
4. an odd index maps to an even index, the round function can be bypassed.

From cases (1) and (3), it follows that round functions are activated whenever non-zero masks propagate to odd indices. As a result, the number of such odd-index positions, denoted by N_o , directly determines the number of active round functions in the trail. The resulting correlation is given by $\mathbf{c}_u = \prod_{i=1}^{(N_o)} \mathbf{c}_i$. Using the methodology from Section 3.2, the expected sum of squared correlations can be expressed as $\mathbb{E} \left(\sum_{u \in \mathbb{F}_2^n \setminus \{0\}} \mathbf{c}_u^2 \right) \approx 2^{n-n(N_o)}$, which leads to the data complexity $q = \mathcal{O}(2^{n(N_o)-n})$.

In the special case where the total number of rounds r is a multiple of the branch number t , the trail evenly alternates between even and odd indices since the order of the branch permutation is t , yielding $N_o = \lfloor r/2 \rfloor$. Consequently, the data complexity of kt rounds distinguishers can be obtained as $q = \mathcal{O}(2^{n\lfloor r/2 \rfloor - n})$. □

For Type 2 GFC $_{\pi}$, where π is a non-even-odd branch permutation, the data complexity of the multidimensional attack depends on the number of odd indices encountered along the trail. When π consists of multiple disjoint cycles, the attack should target the cycle containing the minimum number of odd indices, as the data complexity is directly calculated by the number of odd positions N_o within that cycle.

5 Generic Key Recovery Attacks

The generic multidimensional distinguishers can be used to mount a generic multidimensional key-recovery attack. Hence, we can use the data-complexity results for the r -round multidimensional distinguishers to obtain a $r + r'$ -round key-recovery attack on GFC, utilizing key addition at the input of non-invertible round functions, as seen in Table 8.

Expressing the computation as a matrix–vector product enables an efficient extension to the multidimensional setting. In our case, we have many mask values and can apply the same approach to every approximation. However, the cost will be higher because it essentially involves multiple vectors. By using the corresponding definitions of M and w in Equation (7) and Equation (8), and applying the FFT to them, all masks can be processed simultaneously. As a result, the overall cost remains the same because the test statistic is still a scalar function.

5.1 Type 1 and Type 2 GFC

To improve the work effort of the naive key-recovery method, we apply a variant of the Fast Fourier Transform method [CSQ07] for key-recovery attack on GFC. Suppose we prepend one round to the Type 1 distinguisher. Fix the first branch of the input to a constant c . The LLR statistic for the key-dependent constant c_k is equal to

$$\begin{aligned} \lambda_{c_k} &= \sum_{i=1}^q l(x_{i1} + y_{i0} + c_k) = \sum_{i=1}^q \sum_{z \in \mathbb{F}_2^n} l(x_{i1} + y_{i0} + c_k) \delta_{z, x_{i1} + y_{i0}} \\ &= \sum_{z \in \mathbb{F}_2^n} l(z + c_k) \sum_{i=1}^q \delta_{z, x_{i1} + y_{i0}} = \sum_{z \in \mathbb{F}_2^n} M_{c_k, z} w_z, \end{aligned} \quad (7)$$

where w is a vector of occurrences of $z = x_{i1} + y_{i0}$ values and M is a matrix with entries $M_{c_k, z} = l(z + c_k)$. Here and in the rest of this section, l is the logarithmic likelihood ratio function. The time complexity of computing the matrix M is $\mathcal{O}(2^{2n})$. We exploit the circulant structure of M using the FFT method to reduce the time complexity. Since M is a circulant matrix, it is enough to compute $M_{0, z}$. Even though the vector w and the matrix M differ for other GFC, the calculation of λ is done in the same way. Indeed, we compute the matrix-vector product by taking the pointwise product of the FFT of the first row of M and the FFT of w , followed by applying the inverse Fourier transform. Then the complexity of calculating matrix M becomes $\mathcal{O}(n2^n)$. The complexity of computing the vector w is $\mathcal{O}(2^n + q)$. Thus the total complexity becomes $\mathcal{O}(n2^n + q)$, corresponding to $\mathcal{O}(n2^n)$ when $q \leq n2^n$.

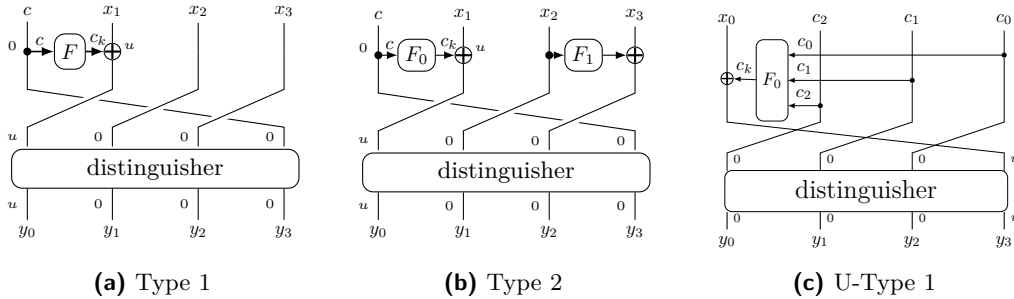


Figure 5: Key recovery attack with FFT method on Type1, Type 2 and U-Type 1 GFC.

The FFT key-recovery method is applied to Type 2, as with Type 1. As seen in Figure 5b, the nonzero mask positions are the same in Type 1 and Type 2. Hence, w is a vector of

occurrences of $z = x_{i1} + y_{i0}$ values and M is a matrix with entries $M_{c_k, z} = l(z + c_k)$ for Type 1 and Type 2. Then to obtain λ_{c_k} , we compute the matrix-vector product such that $\lambda_{c_k} = \sum_{z \in \mathbb{F}_2^n} M_{c_k, z} w_z$.

5.2 U-Type 1 GFC

In U-Type 1 GFC, the round function takes the first three branches as inputs, as seen in Figure 5c. Assuming adding one round to the U-Type 1 distinguisher, set the first three branches of the input to constants c_0, c_1 and c_2 such that $c = c_0 \parallel c_1 \parallel c_2$. The LLR statistic for the key-dependent constant is then given by

$$\lambda_{c_k}^{UT_1} = \sum_{z \in \mathbb{F}_2^n} l(z + c_k) \sum_{i=1}^q \delta_{z, x_{i0} + y_{i3}} = \sum_{z \in \mathbb{F}_2^n} M_{c_k, z} w_z, \quad (8)$$

where w is a vector of occurrences of $z = x_{i0} + y_{i3}$ values and M is a matrix with entries $M_{c_k, z} = l(z + c_k)$. After obtaining the vector w and matrix M , $\lambda_{c_k}^{UT_1}$ can be computed by matrix-vector product using FFT, as described in Section 5.1.

5.3 Recovering the round keys

The attack proceeds by recovering the round key of the active round function while keeping the false-positive rate low, so that the number of key candidates is $K = P_F 2^\ell$ with $P_F \approx 1/2^\ell$ where ℓ is the key bits in the round function. Then, the time complexity becomes $\mathcal{O}(n2^n + q + 2^\ell)$. Extending the distinguisher by r' additional rounds requires repeating these operations for each key guess, resulting in an overall time complexity of $\mathcal{O}(n2^{n(r'-1)\ell} + q + K2^\ell)$ for an $r + r'$ -round key-recovery attack. If $\ell = n$, the time complexity is simplified to $\mathcal{O}(n2^{nr'} + q)$.

5.4 Calculating the maximum number of rounds for key recovery

Multidimensional LLR distinguishers with a data complexity at most $2^{(t-1)n}$ can be extended to generic key recovery attacks for t -branch GFC. Using r -round distinguishers, we can achieve $r + r'$ -round key recovery attacks, where $r = t^2 + t - 1$, and $2t + 1$ with data complexity $2^{(t-1)n}$ for (U)Type 1, and Type 2 GFC, respectively. The value of r' depends on the key size. For a key size of tn , the upper bound on the time complexity is 2^{tn} . Given that the time complexity of an $r + r'$ -round key recovery attack is $n2^{nr'}$, we require $n2^{nr'} < 2^{tn}$, which leads to $r' = t - 1$. Thus, the maximum number of rounds r_k for key recovery attacks is $t^2 + 2t - 2$, and $3t$ for (U)Type 1, and Type 2, respectively, when the key size is 2^{tn} . Similarly, for a key size of 2^{2tn} and $r' = 2t - 1$, r_k increases to $t^2 + 3t - 2$, and $4t$, for (U)Type 1, and Type 2, respectively. Table 8 and Table 9 in Appendix D further validate our calculations. Note that the data complexity of distinguishers is independent of the choice of branch permutations; consequently, modifying the branch permutation does not affect the overall time complexity.

6 Results

In this section, we summarize the maximum number of rounds for generic attacks on t -branch GFC in the KPA model. Moreover, we demonstrate the application of our methods to CAST-256 and GMiM_{crrf}, including comparisons with existing work.

Table 3: Maximum number of rounds r_d and r_k for multidimensional distinguisher and key recovery, for t branches GFC with input size tn and key size k .

Attack type	k	GFC	r_m/r_k
Distinguisher		(U)-Type 1	$t^2 + 2t - 1$
		Type 2	$2t + 3$
Key recovery	tn	(U)-Type 1	$t^2 + 2t - 2$
		Type 2	$3t$
	$2tn$	(U)-Type 1	$t^2 + 3t - 2$
		Type 2	$4t$

6.1 Comparison of the maximum number of rounds for generic attacks

In Table 3, we summarize the maximum number of rounds r_d and r_k for distinguishing and key recovery attacks on GFC with block size tn and key size k . Note that our data complexities are the same for GFC constructed with different branch permutations, as proved in Section 4, and consequently, the values of r_d and r_k are identical regardless of the specific branch permutation used. While improved data complexity may be achievable for Type 2 attacks using non-even-odd branch permutations, such results are not easily generalizable. Therefore, our results in Table 3 for Type 2 are based on an even-odd branch permutation. Firstly, we compare r_d for various t -branch GFC in the KPA model. For Type 1 structures, [NVP13] reports that r_d is $t^2 + t - 2$, while [THW17] improves this to $t^2 + 2t - 2$. Our approach achieves $t^2 + 2t - 1$ rounds, outperforming these results. For Type 2 structures, [NVP13] reports $2t + 2$ rounds, whereas our method extends this to $2t + 3$. For U-Type 1 structures, [PNB06] identifies $2t$ rounds when $t \geq 4$, requiring additional $2^{(2t-4)n}$ computations for higher number of rounds. Our method significantly extends this to $t^2 + 2t - 1$ rounds.

Some prior works, [NVP13, THW17, PNB06], adopt a data complexity limit of 2^{tn} to determine r_d . Accordingly, we follow the same threshold in our analysis to ensure a fair comparison. In contrast, Beyne et al. [BL22] use a stricter limit of $2^{(t-1)n}$ based on truncated differential attacks in the chosen-ciphertext model, achieving up to $t^2 + t - 2$ rounds. Under this constraint, our method achieves $t^2 + t - 1$ rounds, outperforming [BL22]. Since the results are based on the chosen-ciphertext attack in [BL22], these are not included in Table 1. Similarly, the designers of GMiMC [AGP⁺19] provide a security analysis in their paper. However, their results are mostly based on p , which makes a fair comparison with our results difficult.

Due to the absence of a generic key-recovery attack for GFC, we provide a comparison specifically for U-Type 1 structure [BL22, GJNS17]. In [BL22], r_k is $t^2 + 1$, whereas in [GJNS17], it is $5t - 4$; by contrast, our method achieves $t^2 + 2t - 2$ rounds for $k = 4n$. Similarly, for $k = 8n$, our approach achieves $t^2 + 3t - 2$ rounds, surpassing the $7t - 4$ rounds reported in [GJNS17]. These comparisons show that our approach demonstrates significant advancements for Type 1, Type 2, and U-Type 1 GFC/GFC $_{\pi}$.

6.2 Applications of distinguishing attack

This section provides details on the LLR distinguishers for reduced-round CAST-256 and GMiMC_{crf}[$4n, 4, n$]. For 4 branches GFC, data complexities for r -round distinguishers are summarized in Section 3.3, with the explicit formula for the r -round LLR multidimensional distinguisher provided in Equation (5).

For a block size of 128 bits, the expected data complexities are as follows:

- For $4 + d$, $8 + d$, $12 + d$, and $16 + d$ rounds of 4-branch Type 1 and U-Type 1 GFC, the values approximate constant, 2^{32} , 2^{64} , and 2^{96} , respectively, where $d \in \{1, 2, 3\}$.

- For 2 – 3, 4 – 5, 6 – 7, and 8 – 9 rounds of 4-branch Type 2 GFC, the expected complexities similarly approximate constant, 2^{32} , 2^{64} , and 2^{96} .

Table 4: Capacity and approximate data-complexity q for the r -round LLR distinguishers of CAST-256 and GMiMC.

Cipher	r	Round func.	Cap	q
CAST-256	4-7	F_1	1	1
		F_2	1	1
		F_3	1	1
	8-11	F_1, F_2	2.32×10^{-10}	$2^{31.99}$
		F_1, F_3	2.33×10^{-10}	$2^{31.99}$
		F_2, F_3	2.33×10^{-10}	$2^{31.99}$
	12-15	F_1, F_2, F_3	5.46×10^{-20}	$2^{63.99}$
GMiMC	4-7	F	1	1
	8-11	F, F	2.46×10^{-10}	$2^{31.92}$
	12-15	F, F, F	7.41×10^{-20}	$2^{63.55}$

These theoretical predictions are validated experimentally, as presented in Table 4. To efficiently compute the sum of squared correlations (capacity), we use the FFT. Initially, the active round functions encrypt all 2^n possible input values, and the frequency of their output values is stored in different vectors. The FFT is applied to these vectors, and the sum of squared correlations is calculated by summing the squares of the pointwise product of the transformed vectors. For one active round function, the distribution can be computed without FFT; however, computing the pointwise product of Fourier transformations (equivalently, convolution of probability mass functions) becomes a convenient approach for multiple active round functions.

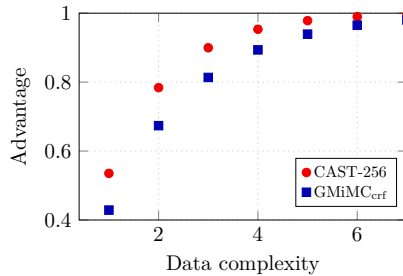


Figure 6: Advantage as a function of data-complexity for 7 rounds LLR distinguishers.

In Table 1, we compare our theoretical estimates of data complexity with those of [NVP13, PNB06, THW17] for r -round distinguishers of GFC with 4 branches. More precise estimates of data complexity can be obtained experimentally by considering the false-positive rate and success probability. In Figure 6, we illustrate the experimental results about the advantage of reduced-round target algorithms with one active round function as a function of data complexity. A direct proportionality is observed between the advantage and the corresponding data complexities.

A summary of experimental data, including the complexities of the reduced-round multidimensional distinguishers with two active round functions for CAST-256 and GMiMC, is presented in Table 5. These results also support our theoretical findings.

In Table 6, we compare our results on linear distinguishers with those of existing works, which show that our approach achieves improved data complexities for most cases, except for the 26-round distinguisher. Using our method, the maximum number of rounds for the distinguishing attack is 23 based on theoretical findings.

Table 5: Data-complexities of LLR multidimensional distinguishers of r -round CAST-256 and GMiMC_{crf} for $P_S \approx 1$ and $P_F \approx 0$.

Cipher	r	Data
CAST-256	4-7	$2^{2.84}$
	8-11	$2^{32.2}$
GMiMC _{crf}	4-7	$2^{2.81}$
	8-11	2^{33}

For GMiMC, we were unable to find results on linear distinguishers in the literature. In [AGP⁺19], the designers of GMiMC evaluate the security of GMiMC_{crf} against linear attacks, concluding that linear attacks pose no threat to the GMiMC family of block ciphers instantiated with the same number of rounds previously defined for differential cryptanalysis without giving details.

Table 6: Comparison of data-complexities q and time-complexities T for LLR distinguishers of r -round CAST-256.

Cipher	r	q	T	Ref.
CAST-256	4	2^3 2^{37}	2^3 2^{37}	§ 6.2 [JR07]
	8	$2^{32.2}$ 2^{69}	$2^{32.2}$ 2^{69}	§ 6.2 [JR07]
	12-15	2^{64} 2^{101}	2^{64} 2^{101}	§ 6.2 [JR07]
	26	$2^{123.7}$	$2^{126.8}$	[ZWW14]

6.3 Application of key recovery attack

We use the FFT method for reduced-round key-recovery attacks. In Section 5, it is already shown that the time complexity of $r + r'$ -round key recovery attack is $\mathcal{O}(n2^{n(r')} + q)$, where q is the data complexity of r -round distinguishers, the round key size is n -bit and $P_F \approx 1/2^n$. For CAST-256 and GMiMC_{crf}[$4n, 4, n$], the block size is 128 bits, consisting of four 32-bit branches, with the subkey size being 32 bits ($n = 32$).

In Table 7, we compare our results on CAST-256 with [WWH09, ZWW14]. For an 18-round key recovery attack, we employ a 15-round distinguisher ($r' = 3$), yielding a data complexity of $2^{2n} = 2^{64}$. Due to the key-dependent rotation in CAST-256, a $5 + 32 = 37$ -bit subkey must be guessed for each appended round. The FFT method determines the time complexity as $2^{37r'} + q$. Therefore, the time complexity for the 18-round key recovery attack is 2^{111} . Similarly, for a 24-round key recovery attack, a 19-round distinguisher (where $r' = 5$) is used. The data complexity in this case is 2^{96} , with a time complexity of 2^{185} . The maximum number of rounds for key recovery is 25, using a 19-round distinguisher ($r' = 6$), which results in a data complexity of 2^{96} and a time complexity of 2^{222} . Beyond 25 rounds, key recovery attacks on CAST-256 become infeasible due to key-dependent rotations. Although the best existing attacks cannot be surpassed, we achieve improved data complexity for 18- and 24-round key recovery. CAST-256 includes three distinct round functions: F_1 , F_2 , and F_3 . When an additional round is inserted before the distinguisher, active round functions can be changed for the corresponding distinguisher. As shown in Table 4, using different round functions has a negligible effect on data complexity. Existing key recovery attacks on GMiMC_{crf}[128, 4, 32] are compared with our results, as summarized in Table 7. For GMiMC_{crf}[128, 4, 32], where the round keys are 32 bits, the

time complexity is given by $2^{37+32(r'-1)} + q$. For a 16-round key recovery attack, we utilize a 15-round distinguisher with parameters $r' = 1$ and $q = 2^{64}$. Under this setup, the time complexity of the 16-round key-recovery attack is 2^{64} . This result is superior to the 16-round attack, which is based on the method detailed in [GJNS17]. Using the same approach, we extend our analysis to a 17-round key recovery attack on GMiMC_{crf}. Here, we apply the 15-round distinguisher with $r' = 2$, resulting in a data complexity of 2^{64} and a time complexity of 2^{69} . The maximum number of rounds for key recovery attack on GMiMC_{crf}[128, 4, 32] is 22, using a 19-round distinguisher ($r' = 3$), which results in a data complexity of 2^{96} and a time complexity of 2^{101} .

The admissible values of r' depend on the key size. Overall, the computational effort increases approximately linearly with the number of repetitions required for the attack. For more information about the data complexities of key recovery attacks on reduced-round GFC, Table 8 and Table 9 in Appendix D can be reviewed.

Table 7: Key recovery results of $r + r'$ rounds CAST-256 and GMiMC_{crf}.

Cipher	$r + r'$	q	T	Ref.
CAST-256	18	2^{64} $2^{79.22}$	2^{111} $2^{111.98}$	§ 6.3 [WWH09]
	24	2^{96} $2^{124.1}$	2^{185} $2^{156.2}$	§ 6.3 [WWH09]
	25	2^{96}	2^{222}	§ 6.3
	32	$2^{126.8}$	2^{251}	[ZWW14]
GMiMC _{crf}	16	2^{64} 2^{112}	2^{64} 2^{96}	§ 6.3 [GJNS17]
	22	2^{96}	2^{101}	§ 6.3

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A Relation between Type 1, Type 2 and Type 3 GFC

Type 2 and Type 3 Generalized Feistel Ciphers can be expressed using Type 1. Let Rot^s denote the right rotational shift by s branch of input. T_1 , T_2 and T_3 represent the function of one round of Type 1, Type 2 and Type 3 GFC (including swap operation) such that

$$\begin{aligned} T_1(x_0, x_1, x_2, x_3) &= (F(x_0) \oplus x_1), x_2, x_3, x_0 \\ T_2(x_0, x_1, x_2, x_3) &= (F(x_0) \oplus x_1), x_2, (F(x_2) \oplus x_3), x_0 \\ T_3(x_0, x_1, x_2, x_3) &= (F(x_0) \oplus x_1), (F(x_1) \oplus x_2), (F(x_2) \oplus x_3), x_0 \\ T_1^{-3}(x_0, x_1, x_2, x_3) &= x_1, (F(x_1) \oplus x_2), (F(x_2) \oplus x_3), (F(x_3) \oplus x_0) \end{aligned}$$

where F is a round function of GFC. Then we have

$$T_2 = \text{Rot}^2 \circ T_1 \circ \text{Rot}^{-1} \circ T_1$$

This equality can be easily shown

$$\begin{aligned} T_2(x_0, x_1, x_2, x_3) &= \text{Rot}^2 \circ T_1 \circ \text{Rot}^{-1} \circ T_1(x_0, x_1, x_2, x_3) \\ &= \text{Rot}^2 \circ T_1 \circ \text{Rot}^{-1}((F(x_0) \oplus x_1), x_2, x_3, x_0) \\ &= \text{Rot}^2 \circ T_1(x_2, x_3, x_0, (F(x_0) \oplus x_1)) \\ &= \text{Rot}^2((F(x_2) \oplus x_3), x_0, (F(x_0) \oplus x_1), x_2) \\ &= (F(x_0) \oplus x_1), x_2, (F(x_2) \oplus x_3), x_0 \end{aligned}$$

Similarly, Type 3 GFC can be represented by the inverse of 3-round Type 1 GFC (T_1^{-3}) and cyclic rotations

$$T_3 = \text{Rot}^1 \circ T_1^{-3} \circ \text{Rot}^1$$

It can essentially be demonstrated

$$\begin{aligned} T_3(x_0, x_1, x_2, x_3) &= \text{Rot}^{-1} \circ T_1^{-3} \circ \text{Rot}^1(x_0, x_1, x_2, x_3) \\ &= \text{Rot}^{-1} \circ T_1^{-3}(x_3, x_0, x_1, x_2) \\ &= \text{Rot}^{-1}(x_0, (F(x_0) \oplus x_1), (F(x_1) \oplus x_2), (F(x_2) \oplus x_3)) \\ &= ((F(x_0) \oplus x_1), (F(x_1) \oplus x_2), (F(x_2) \oplus x_3)), x_0 \end{aligned}$$

B Asymptotic data complexities

For a successful attack, the data complexity must be sufficiently large to guarantee that $\mu_{\text{real}} - \mu_{\text{ideal}} \gg \sigma_{\text{ideal}}$. However, the values of μ and σ may change depending on the statistical test, resulting in different data complexities. Evaluating this condition in both the known-sign and unknown-sign settings shows that, in either case, the data complexity is inversely proportional to the capacity of the multidimensional linear approximation Λ over the function $F : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$, defined as $\text{Cap}(\Lambda) = \sum_{(u,v) \in \Lambda} (C_{v,u}^F)^2$. The rest of this section aims to clarify data complexities of unknown and known-sign cases given in [Nyb13].

B.1 Unknown-sign case

When the signs of correlations are unknown, precise probability distribution estimation requires guessing, and Vaudenay [Vau96] proposed using the χ^2 test. The asymptotic data complexity of the multidimensional χ^2 distinguisher was initially studied in [Nyb13] and

later refined in [HCN19] (Theorem 6.1, Equation (23)). Using the multidimensional χ^2 statistic, the expected value of the test under the real cipher can be expressed as

$$\mu_{\text{real}} = \mathbb{E}[\chi_{\text{real}}^2] \approx |\Lambda| - 1 + q \sum_{(u,v) \in \Lambda \setminus (0,0)} \mathbb{E}[(C_{v,u}^F)^2],$$

where $C_{v,u}^F$ denotes the correlation of the linear approximation associated with the mask pair (u, v) . Also we have $\mu_{\text{real}} = \mathbb{E}[\chi_{\text{ideal}}^2] = |\Lambda| - 1$ and $\sigma_{\text{ideal}}^2 = 2(|\Lambda| - 1)$, it is obtained $\mathbb{E}[\chi_{\text{real}}^2] - \mathbb{E}[\chi_{\text{ideal}}^2] \gg \sqrt{2(|\Lambda| - 1)}$ using the inequality $\mu_{\text{real}} - \mu_{\text{ideal}} \gg \sigma_{\text{ideal}}$. Consequently, a successful χ^2 -based distinguisher requires $q \sum_{(u,v) \in \Lambda \setminus (0,0)} \mathbb{E}[(C_{v,u}^F)^2] \gg \sqrt{|\Lambda|}$, which yields the data-complexity bound

$$q \gg \frac{\sqrt{|\Lambda|}}{\sum_{(u,v) \in \Lambda \setminus (0,0)} \mathbb{E}[(C_{v,u}^F)^2]}.$$

B.2 Known-sign case

When the signs of the correlations are known, the LLR test proposed by Vaudenay et al. [BJV04] can be used. The asymptotic data complexity of the multidimensional LLR distinguisher was initially studied in [Nyb13] and later refined in [HCN19] (Corollary 6.1, Equation (33)). The test statistic is $\lambda = \sum_{(u,v) \in \Lambda} w_{v,u} \hat{c}_{v,u}$, where each $\hat{c}_{v,u}$ is an empirical estimator of the correlation associated with the mask pair (u, v) . To keep the variance of λ constant, the weights must satisfy $\sum_{(u,v) \in \Lambda} w_{v,u}^2 = 1$. If the samples come from the real cipher, then the expected value is

$$\mathbb{E}[\lambda_{\text{real}}] = \sum_{(u,v) \in \Lambda} w_{v,u} C_{v,u}^F.$$

The weights must be chosen to maximize $\mathbb{E}[\lambda_{\text{real}}]$ while keeping $\sum w^2 = 1$. The optimal choice is $w_{v,u} = \frac{C_{v,u}^F}{\sqrt{\sum_{(u,v) \in \Lambda} (C_{v,u}^F)^2}}$. With these weights, the expected value under the real cipher becomes $\mathbb{E}[\lambda_{\text{real}}] = \sqrt{\sum_{(u,v) \in \Lambda} (C_{v,u}^F)^2}$.

If the samples come from the ideal distribution, then $\mathbb{E}[\lambda_{\text{ideal}}] = 0$ and their average is $1/q$. A successful attack, therefore, requires $\mathbb{E}[\lambda_{\text{real}}] \gg \frac{1}{\sqrt{q}}$, which implies

$$q \gg \frac{1}{\sum_{(u,v) \in \Lambda} (C_{v,u}^F)^2}.$$

C Multidimensional linear attack on Type 3 and U-Type 3 GFC

In Figure 7, U-Type 1 and U-Type 3 GFC are given for 4 branches.

C.1 Data-complexities of generic multidimensional linear distinguisher of r -round and t -branch (U)-Type 3 GFC

The data complexity of the generic multidimensional linear attack of Type 3 and U-Type 3 GFC is calculated as in Section 3.1. In this case, we have

$$\Lambda_{T_3} = \Lambda_{U_3} = \left\{ ((u, 0, \dots, 0, 0), (u, 0, \dots, 0, 0)) \mid u \in \mathbb{F}_2^n \right\}.$$

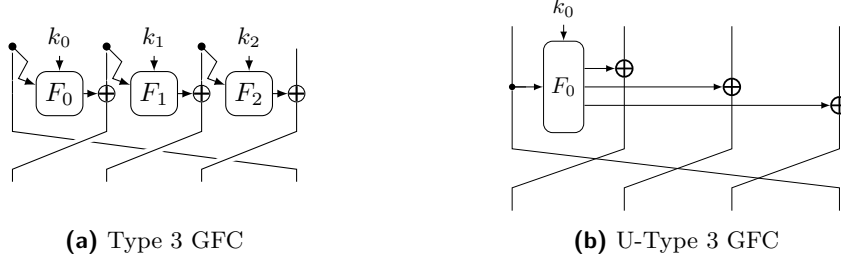


Figure 7: One round of 4-branch (U)-Type 3 GFC.

As can be understood from Figure 8, we can skip the first round, so $t - 1$ round functions are active for every t round for Type 3 and U-Type 3 GFC. We get $\mathbb{E} \left(\sum_{u \in \mathbb{F}_2^n \setminus \{0\}} \mathbf{c}_u^2 \right) \approx 2^{n-n(r-1)}$. Then, the asymptotic data-complexity of the LLR test is $q = \mathcal{O}(2^{n(r-1)-n})$.

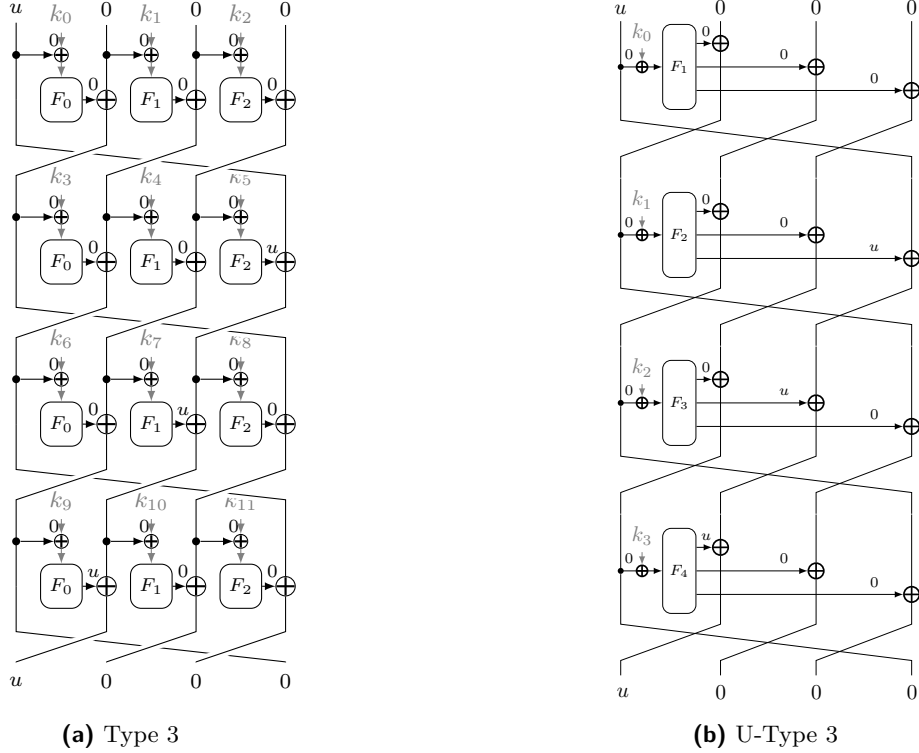


Figure 8: 4-round iterative linear trail for a 4-branch (U)-Type 3 GFC with key-independent correlations.

C.1.1 The maximum number of rounds for distinguishing attack:

Our analysis adopts 2^{tn} as the upper bound for the data complexity threshold since there are t branches with branch-size n . For Type 3 and U-Type 3 GFC, the condition $2^{n(r-1)-n} \leq 2^{tn}$ results in $r \leq t + 2$. For Type 3 structures, the maximum number of rounds is $3t/2 + 1$ in [NVP13] and $t + 2$ in [THW17], which matches our result of $t + 2$. In [VNM17], the authors mentioned improvements over the results in [NVP13]. However, the authors did not provide precise data complexity depending on n for their attacks and

stated that their attack is more effective for smaller values of n (for example, $n = 8$). This lack of specificity makes direct comparison with our results challenging. In the case of U-Type 3 structures, [PNB07] reports a maximum of $3t$ rounds for $t \geq 3$, requiring additional permutations or 2^{tn} computations for further rounds, while [VNP10] achieves $3t - 1$ rounds with $2^{(t-1/2t+2)n}$ data. Our approach achieves $t + 2$ rounds, which is less effective than existing results for U-Type 3 structures. Our approach demonstrates that existing results remain superior for (U)-Type 3.

C.1.2 Applications of distinguishing attack on (U)-Type3:

Given samples $z_1, \dots, z_q$, the logarithmic likelihood-ratio (LLR) statistic is equal to

$$\lambda_{\text{real}} = \log \prod_{i=1}^q \frac{p_{\text{real}}(z_i)}{p_{\text{ideal}}(z_i)} = \sum_{i=1}^q \log \frac{p_{\text{real}}(z_i)}{p_{\text{ideal}}(z_i)},$$

Let $(x_0^i, x_1^i, \dots, x_{t-1}^i, x_t^i)$ and $(y_0^i, y_1^i, \dots, y_{t-1}^i, y_t^i)$ be q plaintext and corresponding ciphertext samples respectively, for $i \in \{1, 2, \dots, q\}$. Then, $z_i = x_0^i + y_0^i$ for Type 3, and U-Type 3. As mentioned in Section 2.2, the asymptotic data-complexity of the LLR test is inversely proportional to the sum of squared correlations, then we have $q = \mathcal{O}(2^{(r-1)n-n})$ for Type 3 and U-Type 3 where $r \geq 2$. For a block size of 128 bits, the expected data complexities are For 2, 3, 4, and 5 rounds of Type 3 and U-Type 3 GFC, the expected data complexities are constant, 2^{32} , 2^{64} , and 2^{96} . To experimentally verify these results, we need to follow the attack scenario given in Section 3.3 and Section 6.2 with the obtained z_i values. GMiMC_{eff} is analyzed to demonstrate the applicability of our attack to U-Type 3. Data-complexities of LLR multidimensional distinguishers of 2 and 3 rounds GMiMC_{eff} are $2^{3.1}$ and 2^{33} , respectively, where $P_S \approx 1$ and $P_F \approx 0$. These experimental results validate our theoretical results.

C.2 Generic key recovery attack on Type 3 and U-Type 3 GFC

The generic key recovery attack scenario of Type 3 GFC is similar to Type 1 since nonzero mask positions can be the same. If we fix the first branch of the input to a constant c , the LLR statistic for the key-dependent constant c_k is equal to

$$\begin{aligned} \lambda_{c_k} &= \sum_{i=1}^q l(x_{i1} + y_{i0} + c_k) = \sum_{i=1}^q \sum_{z \in \mathbb{F}_2^n} l(x_{i1} + y_{i0} + c_k) \delta_{z, x_{i1} + y_{i0}} \\ &= \sum_{z \in \mathbb{F}_2^n} l(z + c_k) \sum_{i=1}^q \delta_{z, x_{i1} + y_{i0}} = \sum_{z \in \mathbb{F}_2^n} M_{c_k, z} w_z, \end{aligned} \quad (9)$$

where w is a vector of occurrences of $z = x_{i1} + y_{i0}$ values and M is a matrix with entries $M_{c_k, z} = l(z + c_k)$. To obtain λ_{c_k} , we compute the matrix-vector product such that $\lambda_{c_k} = \sum_{z \in \mathbb{F}_2^n} M_{c_k, z} w_z$. Thus the total complexity becomes $\mathcal{O}(n2^n + q)$, corresponding to $\mathcal{O}(n2^n)$ when $q \leq n2^n$. If we incorporate additional rounds after the distinguisher for the key recovery, the same operations must be performed for each key guess.

In U-Type 3 GFC, the round function takes the first branch as input, and outputs are XORed with the other three branches, as seen in Figure 9b. Similar to the other cases, we prepend one round to the distinguisher and keep the input of the round function constant. Then, we get the output of the round function such that $c_k = c_{0k}, c_{1k}, c_{2k}$. Now, we can calculate the LLR test statistic for the key-dependent constant:

$$\lambda_{c_{0k}}^{UT_3} = \sum_{i=1}^q l(x_{i1} + y_{i0} + c_{0k}) = \sum_{z \in \mathbb{F}_2^n} M_{c_{0k}, z} w_z, \quad (10)$$

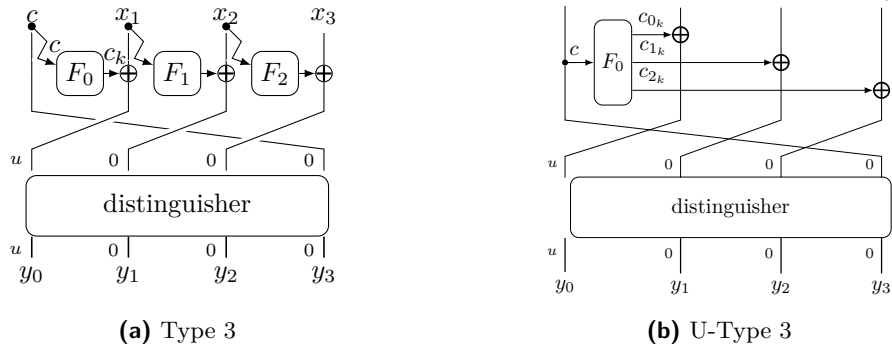


Figure 9: Key recovery with FFT method using (U)-Type 3 distinguishers.

where w is a vector of occurrences of $z = x_{i1} + y_{i0}$ values and M is a matrix with entries $M_{c_{0k}, z} = l(z + c_{0k})$. After obtaining the vector w and matrix M , $\lambda_{c_{0k}}^{UT_3}$ can be computed by matrix-vector product using FFT, as described in in Section 5.1.

C.2.1 The maximum number of rounds for key recovery attack:

Multidimensional LLR distinguishers with a data complexity less than $2^{(t-1)n}$ can be extended to generic key recovery attacks for t -branch GFC. Using r -round distinguishers, we can achieve $r + r'$ -round key recovery attacks, where $t + 1$ for (U)Type 3 structures. The value of r' depends on the key size. For a key size of 2^{tn} , the upper bound on the time complexity is 2^{tn} . Given that the time complexity of an $r + r'$ -round key recovery attack is $n2^{nr'}$, we require $n2^{nr'} < 2^{tn}$, which leads to $r' = t - 1$. Thus, the maximum number of rounds for key recovery is $2t$ for (U)Type 3, when the key size is 2^{tn} . Similarly, for a key size of 2^{2tn} and $r' = 2t - 1$, the maximum number of rounds for key recovery attack increases to $3t$ for (U)Type 3, respectively. Table 8 and Table 9 in Appendix D further validate our calculations. After obtaining the vector w and matrix M , $\lambda_{c_k}^{UT_1}$ can be computed by matrix-vector product using FFT, as described in Section 5.1.

D Figures and tables

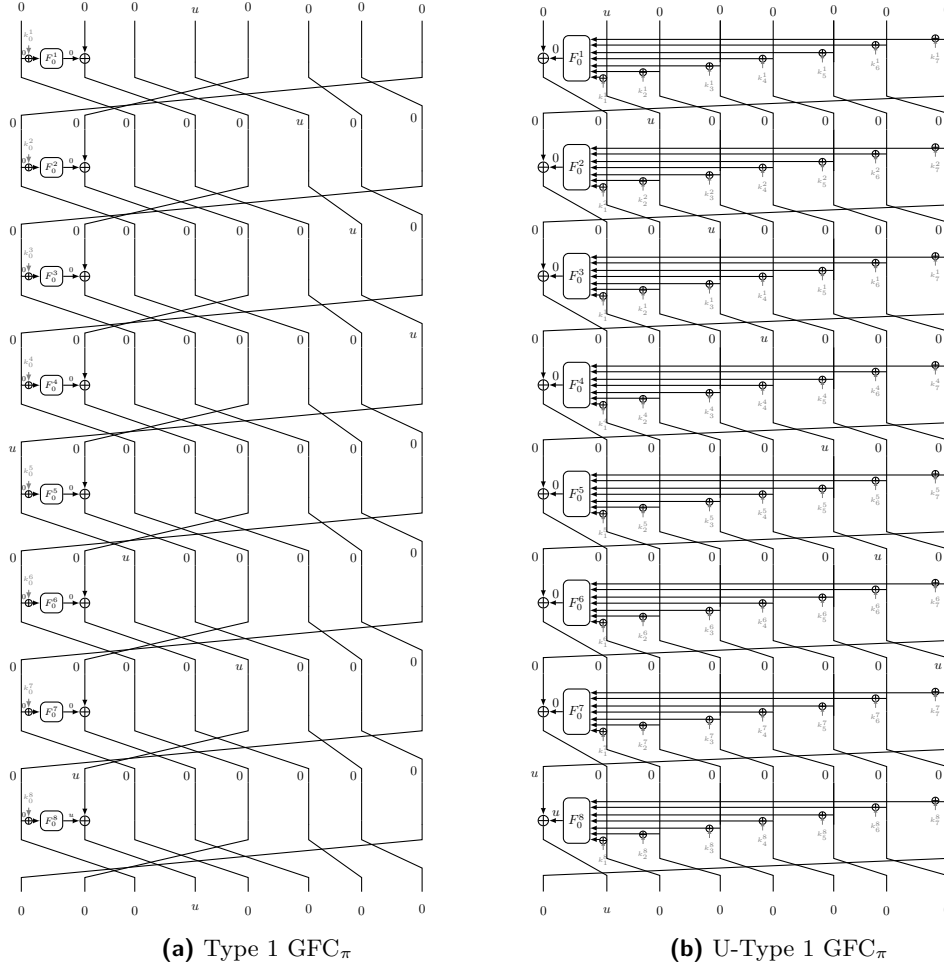


Figure 10: Eight-round linear trail for 8 branches GFC described by Shingo and Iwata [YI11] with key-independent correlation.

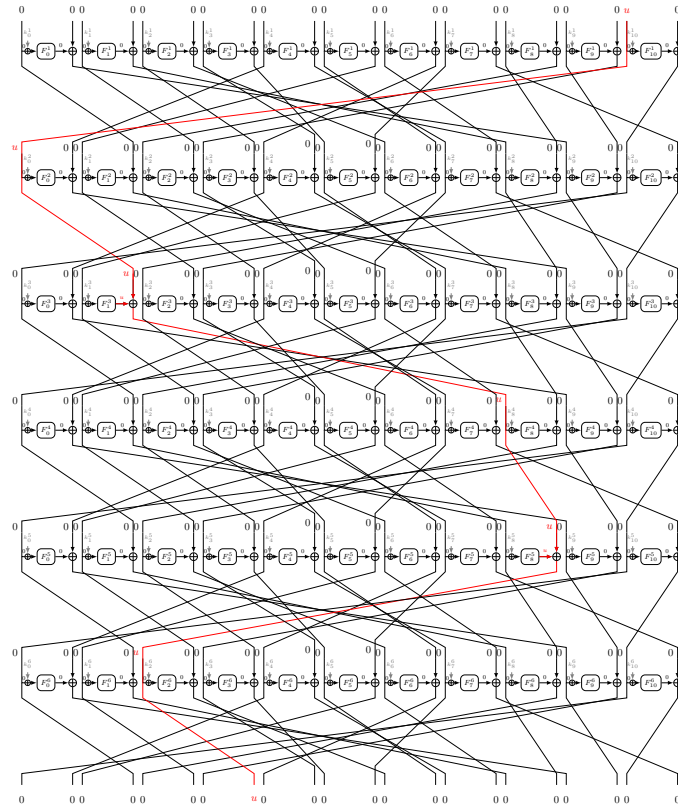


Figure 11: Six-round linear trail for the 22 branches Type 2_{π_1} GFC described by Delaune et al. [DDGP22] with key-independent correlation.

Table 8: Complexities of key recovery attacks on $r + r'$ -round GFC based on r -round distinguishers for 4 branches with block size $4n$ and key length k . All work efforts and data-complexities are given up to constant factors. All distinguishers are in the known-plaintext model.

Types	k	r	$r + r'$	Data	Work effort
Type 1 U-Type 1	4n	7	10	1	$n2^{3n}$
		11	14	2^n	$n2^{3n}$
		15	19	2^{2n}	$n2^{3n}$
		19	22	2^{3n}	$n2^{3n}$
	8n	7	14	1	$n2^{7n}$
		11	18	2^n	$n2^{7n}$
		15	22	2^{2n}	$n2^{7n}$
		19	26	2^{3n}	$n2^{7n}$
Type 2	4n	3	6	1	$n2^{3n}$
		5	8	2^n	$n2^{3n}$
		7	10	2^{2n}	$n2^{3n}$
		9	11	2^{3n}	$n2^{3n}$
	8n	3	10	1	$n2^{7n}$
		5	12	2^n	$n2^{7n}$
		7	14	2^{2n}	$n2^{7n}$
		9	16	2^{3n}	$n2^{7n}$
Type 3 U-Type 3	4n	2	5	1	$n2^{3n}$
		3	6	2^n	$n2^{3n}$
		4	7	2^{2n}	$n2^{3n}$
		5	8	2^{3n}	$n2^{3n}$
	8n	2	9	1	$n2^{7n}$
		3	10	2^n	$n2^{7n}$
		4	11	2^{2n}	$n2^{7n}$
		5	12	2^{3n}	$n2^{7n}$

Table 9: Generic key recovery attack on $r + r'$ rounds 4-branch GFC using r rounds multidimensional LLR distinguisher where the key size is $8n$.

(U)-Type 1				Type 2				(U)-Type 3			
r	$r + r'$	q	T	r	$r + r'$	q	T	r	$r + r'$	q	T
15	16	2^{2n}	2^{2n}	7	8	2^{2n}	2^{2n}	4	6	2^{2n}	2^{2n}
	17	2^{2n}	$n2^{2n}$		9	2^{2n}	$n2^{2n}$		7	2^{2n}	$n2^{2n}$
	18	2^{2n}	$n2^{3n}$		10	2^{2n}	$n2^{3n}$		8	2^{2n}	$n2^{3n}$
	19	2^{2n}	$n2^{4n}$		11	2^{2n}	$n2^{4n}$		9	2^{2n}	$n2^{4n}$
	20	2^{2n}	$n2^{5n}$		12	2^{2n}	$n2^{5n}$		10	2^{2n}	$n2^{5n}$
	21	2^{2n}	$n2^{6n}$		13	2^{2n}	$n2^{6n}$		11	2^{2n}	$n2^{6n}$
	22	2^{2n}	$n2^{7n}$		14	2^{2n}	$n2^{7n}$		12	2^{2n}	$n2^{7n}$
19	20	2^{3n}	2^{3n}	9	10	2^{3n}	2^{3n}	5	7	2^{3n}	2^{3n}
	21	2^{3n}	2^{3n}		11	2^{3n}	2^{3n}		8	2^{3n}	$n2^{3n}$
	22	2^{3n}	$n2^{3n}$		12	2^{3n}	$n2^{3n}$		9	2^{3n}	$n2^{4n}$
	23	2^{3n}	$n2^{4n}$		13	2^{3n}	$n2^{4n}$		10	2^{3n}	$n2^{5n}$
	24	2^{3n}	$n2^{5n}$		14	2^{3n}	$n2^{5n}$		11	2^{3n}	$n2^{6n}$
	25	2^{3n}	$n2^{6n}$		15	2^{3n}	$n2^{6n}$		12	2^{3n}	$n2^{7n}$
	26	2^{3n}	$n2^{7n}$		16	2^{3n}	$n2^{7n}$				